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Light disinfecting apparatus and methods

Abstract

An assembly for redirecting light emitted by an end-emitting optical fiber into an inner lumen of a body is provided. According to one implementation, the body includes one or more surfaces disposed on or in the body onto which the light is configured to impinge when the end emitting optical fiber is activated, the one or more surfaces being configured to alter the trajectory of the light so that the light is directed to impinge on a light reflector of a cap removably attached to the body, the light reflector of the cap being configured to redirect the light distally into the inner lumen of the body.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of application Ser. No. 17/015,831, filed Sep. 9, 2020, which is a continuation application of application Ser. No. 16/863,419, filed Apr. 30, 2020, now U.S. Pat. No. 11,357,875, which is a continuation of application Ser. No. 15/876,796, filed Jan. 22, 2018, now U.S. Pat. No. 10,675,368, which is a continuation-in-part of application Ser. No. 15/631,674, filed Jun. 23, 2017, now U.S. Pat. No. 9,925,287, which is a continuation of application Ser. No. 15/629,494 filed Jun. 21, 2017, now U.S. Pat. No. 9,925,285.

TECHNICAL FIELD

(1) The present disclosure relates to devices having disinfecting capabilities and to methods for disinfecting any of a host of devices including, for example, devices used in the medical treatment of patients.

BACKGROUND

(2) Unwanted and dangerous bacteria growth can occur on or in devices that are commonly used to treat patients. These devices may include central venous catheters, urinary catheters, ventilators, wound protection devices, etc. Hospital acquired infections account for a substantial yearly expense to hospitals and insurance companies, and are a major cause of extending hospital stays for patients. Equipment or components of water processing plants, food processing plants, dairies, livestock habitation facilities, etc. are also susceptible bacteria growth.

SUMMARY OF THE DISCLOSURE

(3) According to some implementations disclosed herein light is used to disinfect the internal and/or external surfaces of devices used in the medical treatment of patients. The light may be any wavelength of light that is capable of killing bacteria, such as, for example, ultra violet (UV) light and blue light which may be delivered by one or both of a radially emitting optical fiber and an end emitting optical fiber.

(4) An advantage of using light to kill bacteria is that it is not susceptible to the danger of antimicrobial resistance that can occur with the use of pharmacologic or chemical agents. Another advantage is that there are severe side effects associated with many pharmacologic or chemical agents are avoided.

(5) According to other implementations light based detection means is used to monitor the formation of clots and/or bacterial biofilms on the outer surface of catheters residing in a vessel or duct of a patient.

(6) It is important to note that although the forthcoming disclosure is directed primarily to medical devices, it is in no way limited to such devices. For example, the apparatus and methods disclosed herein related to killing bacteria with light and the monitoring of clot or biofilm growth can also be applied to equipment or components of water processing plants, food processing plants, dairies, livestock habitation facilities, etc.

(7) These and other advantages and features will become evident in view of the drawings and detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A and 1B respectively show a side view and cross-section view of a radially emitting optical fiber according to some implementation;

(2) FIG. 2 is a perspective view of an end emitting optical fiber according to some implementations;

(3) FIG. 3 is a perspective view of a central venous catheter assembly according to some implementations.

(4) FIG. 4A is a cross-section view of a main shaft of a central venous catheter according to one implementation.

(5) FIG. 4B is a perspective view of the main shaft depicted in FIG. 4A.

(6) FIG. 4C is a perspective view of a distal end portion of the main shaft depicted in FIGS. 4A and 4B.

(7) FIG. 4D is a perspective view of a proximal end portion of the main shaft according to one implementation.

(8) FIGS. 5A and 5B show cross-section views of a main shaft of a central venous catheter according to other implementations.

(9) FIG. 6 shows a distal end portion of the main shaft having a spiral imaging fiber located adjacent apertures in the main shaft that extend to an exterior surface of the main shaft.

(10) FIGS. 7A and 7B illustrate an imaging system associated with a main shaft of a central venous catheter according to one implementation.

(11) FIG. 8A is an external perspective view of an infusion shaft according to one implementation.

(12) FIG. 8B is an exploded view of the infusion shaft of FIG. 8A.

(13) FIGS. 9A and 9B are cross-section views of a tubular body of an infusion shaft according to one implementation.

(14) FIGS. 9C through 9F show alternative exemplary implementations of an infusion shaft having reflectors that guide the light emitted by the radially emitting fiber into the working lumen.

(15) FIGS. **10A-C** illustrate cross-section views of the main shaft according to other implementations.

(16) FIG. **11A** illustrates an implementation of a clamp used to close off flow through an infusion shaft with the clamp being in the open position.

(17) FIG. **11B** illustrates the clamp of FIG. **11A** being in the closed position.

(18) FIG. **11C** illustrates the infusion shaft running through the clamp of FIG. **11A** with the clamp being in the open position.

(19) FIG. **12** is a cross-section side view of a tubular body of an infusion shaft according to one implementation.

(20) FIG. **13** is a perspective view of the proximal end portion of the tubular body of FIG. **12**

(21) FIG. **14A** is a cross-section side view of an infusion shaft proximal connector according to one implementation.

(22) FIG. **14B** is a cross-section view showing the proximal end segment of the key portion of the tubular body inserted into the receiving lumen of the proximal connector.

(23) FIG. **15** illustrates a path of the light emitted by the end emitting fiber through the proximal connector of FIG. **13**.

(24) FIG. **16** is an external perspective view of an infusion shaft proximal connector according to one implementation.

(25) FIG. **17** is an internal view of the guidewire ramp and optical surface of FIG. **13**.

(26) FIG. **18** is a perspective view of the cap of FIG. **13**.

(27) FIGS. **19A-D** illustrate cross-section side views of a portions of an infusion shaft according to other implementations.

(28) FIG. **20A** is a cross-section side view of an infusion shaft proximal connector according to one implementation.

(29) FIG. **20B** illustrates a path of the light emitted by the end emitting fiber through the proximal connector of FIG. **20A**.

(30) FIG. **21A** is a cross-section side view of an infusion shaft proximal connector according to one implementation.

(31) FIG. **21B** illustrates a path of the light emitted by the end emitting fiber through the proximal connector of FIG. **21A**.

(32) FIG. **22A** illustrates a laser system that delivers light from a single laser to eight optical fibers of a central venous catheter according to one implementation.

(33) FIG. **22B** illustrates a laser system that delivers light from eight lasers to a respective eight optical fibers of a central venous catheter according to one implementation.

(34) FIG. **22C** illustrates a laser system that delivers light from three lasers to eight optical fibers of a central venous catheter according to one implementation.

(35) FIG. **23** shows a top view of an optical umbilical cord of a central venous catheter according to one implementation.

(36) FIG. **24A** shows a perspective view of a distal end of the umbilical cord of FIG. **23** according to one implementation.

(37) FIG. **24B** is an exploded view of the component parts of the optical fibers shown in FIG. **24A**.

(38) FIG. **25** show top perspective views of the components of a hub according to one implementation, the components including a bottom tray, a mid-tray, a cover and a gasket.

(39) FIG. **26** is an enlarged view of the bottom tray of FIG. **25**.

(40) FIGS. **27A** and **27B** are enlarged perspective views of the top and bottom of the mid-tray of FIG. **25**, respectively.

(41) FIG. **28** is an enlarged view of the cover of FIG. **25**.

(42) FIG. **29** is an enlarged view of the gasket of FIG. **25**.

(43) FIG. **30** is a perspective view of a partially assembled hub according to one implementation.

(44) FIG. **31** is a perspective view of the hub of FIG. **30** with a cover positioned to protect the

optical fibers running through the hub.

(45) FIG. 32 shows an elongate beading used to form a conduit within the hub.

(46) FIG. 33 is a perspective view of a partially assembled hub according to another implementation.

(47) FIG. 34 is a perspective view of a central venous catheter with the hub containing one or more light reflectors on its outer surface.

(48) FIG. 35A is a top perspective view of a partially assembled hub according to one implementation.

(49) FIG. 35B shows the hub of FIG. 35A without the optical fibers.

(50) FIG. 36 is a top view of the hub shown in FIG. 35A.

(51) FIG. 37 is a top perspective view of a mid-tray according to one implementation.

(52) FIG. 38 is a cross-section side view of the hub shown in FIG. 35A.

(53) FIGS. 39A-D are perspective views of a partially constructed hub of a central venous catheter according to other implementations.

(54) FIG. 40 illustrates a partially constructed hub of a central venous catheter according to another implementation.

(55) FIG. 41 shows a fixture for routing optical fibers within the hub of FIG. 40

(56) FIG. 42 is a perspective view of a partially constructed hub of a central venous catheter according to another implementation.

(57) FIG. 43 is a transparent view of a manifold of the hub of FIG. 43 having internal conduits that are fluidly coupled to the working lumens of the main shaft.

(58) FIG. 44 is a perspective view of the distal side of the manifold of FIG. 43.

(59) FIGS. 45A and 45B show perspective views of a manifold adapter according to one implementation.

(60) FIG. 46 shows mandrels being supported inside an infusion shaft support fixture and the manifold adapter of the hub of FIG. 42.

(61) FIG. 47A shows a proximal side of the infusion shaft support fixture of FIG. 46.

(62) FIG. 47B shows a distal side of the infusion shaft support fixture of FIG. 46.

(63) FIG. 48A is a perspective side view of an end emitting fiber having an end cap.

(64) FIG. 48B is a cross-section view of the end emitting fiber of FIG. 48A.

DETAILED DESCRIPTION

(65) FIG. 1A is a schematic side view of a radially emitting fiber with a plurality of voids in the core of the radially emitting optical fiber 12 having a central axis 16. FIG. 1B is a schematic cross-section of a radially emitting optical fiber 12 as viewed along the direction 1B-1B in FIG. 1A. Radially emitting fiber 12 can be, for example, an optical fiber with a nano-structured fiber region having periodic or non-periodic nano-sized structures 32 (for example voids). In an example implementation, fiber 12 includes a core 20 divided into three sections or regions. These core regions are: a solid central portion 22, a nano-structured ring portion (inner annular core region) 26, and outer, solid portion 28 surrounding the inner annular core region 26. A cladding region 40 surrounds the annular core 20 and has an outer surface. The cladding 40 may have low refractive index to provide a high numerical aperture. The cladding 40 can be, for example, a low index polymer such as UV or thermally curable fluoroacrylate or silicone.

(66) An optional coating 44 surrounds the cladding 40. Coating 44 may include a low modulus primary coating layer and a high modulus secondary coating layer. In at least some implementations, coating layer 44 comprises a polymer coating such as an acrylate-based or silicone based polymer. In at least some implementations, the coating has a constant diameter along the length of the fiber.

(67) In other exemplary implementations, coating 44 is designed to enhance the distribution and/or the nature of radiated light that passes from core 20 through cladding 40. The outer surface of the cladding 40 or the of the outer of optional coating 44 represents the sides 48 of fiber 12 through

which light traveling in the fiber is made to exit via scattering, as described herein.

(68) A protective jacket (not shown) optionally covers the cladding **40**.

(69) In some implementations, the core region **26** of radially emitting fiber **12** comprises a glass matrix **31** with a plurality of non-periodically disposed nano-sized structures (e.g., voids) **32** situated therein, such as the example voids shown in detail in the magnified inset of FIG. **1B**. In another example implementation, voids **32** may be periodically disposed, such as in a photonic crystal optical fiber, wherein the voids may have diameters between about 1×10^{-6} m and 1×10^{-5} m. Voids **32** may also be non-periodically or randomly disposed. In some exemplary implementations, glass **31** in region **26** is fluorine-doped silica, while in other implementations the glass may be an undoped pure silica.

(70) The nano-sized structures **32** scatter the light away from the core **20** and toward the outer surface of the fiber. The scattered light is then diffused through the outer surface of the fiber **12** to provide the desired illumination. That is, most of the light is diffused (via scattering) through the sides of the fiber **12** and along the fiber length without the need to remove any portion of the cladding **40**.

(71) According to some implementations the nano-sized structures **32** are formed in the cladding **40** of the fiber in lieu of or in conjunction with providing nano-sized structures in the core **12**.

(72) According to some implementations the core **20** has a diameter in the range of 125-300 μm and the overall diameter of the fiber system, including the protective jacket, is in the range of 700 to 1200 μm . According to some implementation, the outer diameter of the fiber **12** without a jacket is in the range of 200-350 μm .

(73) A detailed description of exemplary radially emitting optical fibers may be found in Reissue Pat. No. RE46,098 whose content is incorporated herein by reference in its entirety.

(74) An example of a radially emitting optical fiber is the Fibrance® Light Diffusing Fiber manufactured by Corning® Incorporated located in Corning, New York. The Fibrance® Light Diffusing Fiber has many of the attributes of the radially emitting fiber **12** described above. An advantage of the Fibrance® Light Diffusing Fiber is that it emits light essentially along its entire length and has a small functional bend radius of around 5 millimeters which allows it be easily bent to assume a host of shapes. Breakage of the fiber typically occurs when it is bent to a bend radius of less than about 2 millimeters.

(75) Radially emitting fibers like those disclosed in Reissue Pat. No. RE46,908 do not require the removal of a light reflective component or light reflective element to enable the emission of light radially from the optical fiber.

(76) An end emitting optical fiber is an optical fiber that emits light from a terminal end of the fiber. Such emitted light is referred to herein as “end emitted light” A multimode optical fiber like that shown in FIG. **2**, is one example of an end emitting optical fiber wherein light is guided down the center of the fiber through the core **51** and out the end thereof. The fiber **50** includes a core **51** surrounded by a cladding **52**. The cladding **52** has a lower index of refraction than the core **51** and traps the light in the core using an optical technique called “total internal reflection.” The fiber **50** itself may include a coated “buffer” to protect the fiber from moisture and physical damage. The core **51** and cladding **52** are usually made of ultra-pure glass, although some fibers are all plastic or a glass core and plastic cladding. According to some implementations the core **51** has a diameter in the range of 50-250 μm and the diameter of the cladding **52** is typically around 100-500 μm . The overall diameter of the fiber system, including the buffer coating **53**, is typically around 150-750 μm . Breakage of the fiber typically occurs when it is bent to a bend radius of less than about 2 millimeters.

(77) A “transport fiber” as used herein, refers to an optical fiber that transports light longitudinally through its core to an end of the fiber with little loss. That is, the vast majority (e.g., $\geq 90\%$) of the light fed into a proximal end of the transport fiber is delivered to the terminal end of the fiber. As explained in more detail below, transport fibers are used in a variety of the implementations

disclosed and contemplated herein to couple a light source (e.g., a laser) to a radially emitting optical fiber and/or end emitting fiber. According to some implementations, the transport fibers disclosed herein are multimode optical fibers.

(78) It is important to note that a radially emitting optical fiber, like the examples discussed above, may also emit light from the core **20** at a terminal end of the radially emitting optical fiber **12**. Thus, according to some implementations a disinfecting of a device may occur as a result of bacterial disinfecting light being emitted from both the circumference and the end of a radially emitting fiber. An optical fibers designated for this use is referred to herein as a “dual emitting fiber”.

(79) Blue light and ultra-violet light have been shown to kill or curtail the growth of certain types of unwanted bacteria that is hazardous and potentially fatal to mammalian life. Examples of such bacteria are *Staphylococcus aureus*, *Pseudomonas aeruginosa*, *Leuconostoc mesenteroides*, *Bacillus atrophaeus*, *Escherichia coli*, Coagulase-negative staphylococci etc. In treatments involving a mammal, blue light is preferred over ultra-violet light due to detrimental effects of ultra-violet light on mammalian cells and possible damage to host tissue. In accordance with some implementations disclosed herein blue light at a wavelength of between 380-495 nm and an exposure of between 100-1,000 Joules/cm² is employed to kill the unwanted bacteria. According to other implementations, ultra-violet light at a wavelength of 100-400 nm and exposure up to 6 J/cm² is employed to kill unwanted bacteria.

(80) It is important to note that the present disclosure is in no way limited to the use of blue light and ultra-violet light to kill unwanted bacteria. As briefly explained above, the present disclosure contemplates the use of any type of light that is susceptible to killing unwanted bacteria.

(81) FIG. 3 illustrates a central venous catheter (CVC) **100** that will serve as an exemplary device in which light emitting optical fibers are used to disinfect one or both of the internal and external surfaces of the device. It will become evident in the forthcoming disclosure that many of the same principles employed in the CVC example are equally applicable to a host of other medical and non-medical devices.

(82) A CVC, also called a central line, a long, thin, polymeric, flexible tube (referred to herein as the “main shaft”) used to deliver medicines, fluids, nutrients, or blood products to a venous system of a patient over a long period of time, usually from days up to several weeks or more. The main shaft is often inserted into the arm or chest through the skin into a vein. The main shaft is typically threaded through this vein until it reaches a larger vein, such as, for example, a large vein near the heart.

(83) According to some implementations one or more or all of the light emitting optical fibers that pass through the CVC are unjacketed optical fibers that include only a core and a cladding. This reduces the diametric profile of the optical fibers that allows them to be more readily integrated into a CVC, or other types of devices, without substantially altering the manner in which the CVC is traditionally used. In this way, established clinical practices may be followed. The lower diametric profile also beneficially enables the parts of the CVC to be scaled to a smaller size.

(84) FIG. 3 depicts a perspective view of a CVC **100** according to one implementation. In this implementation the CVC includes a main shaft **200** having three working lumens (see FIGS. 4A, 5A and 5B) through which different types of therapeutic agents may be delivered to the patient. The working lumens may also serve as conduits for receiving other types of medical instruments such as, for example, a guidewire that is used to guide the distal end portion of the main shaft **200** to a desired location in the venous system.

(85) In the example of FIG. 3, the CVC **100** includes three infusion shafts **300** having working lumens **308** that are fluidly and respectively coupled to three working lumens **201a**, **201b** and **201c** of the main shaft **200** through a hub **400**. That is, the lumen **308** of each of the infusion shafts **300** is separately fluidly coupled to one of the working lumens of the main shaft **200**. The main shaft **200** of the CVC **100** may comprise more or fewer working lumens with there being a

corresponding number of infusion shafts. For example, the main shaft **200** of the CVC **100** may have one, two or four working lumens with a corresponding one, two or four infusion shafts **300**. (86) A light delivery umbilical **500** comprising one or more transport fibers may be provided to transport light from a light source to one or more optical fibers disposed in one or more of the main shaft **200**, infusion shafts **300** and hub **400**. The light delivery umbilical **500** may include one or more proximal connectors **501** to couple one or more light sources to the one or more transport fibers.

(87) What follows is a detailed description of exemplary implementations of each of the main shaft **200**, infusion shafts **300**, hub **400**, light delivery umbilical **500** and laser system.

(88) FIG. 4A illustrates a cross-sectional view of a main shaft **200** of a CVC according to one implementation. As discussed above, according to some implementations the main shaft **200** includes first, second and third working lumens **201a**, **201b** and **201c**, respectively. According to some implementations, as shown in FIG. 4B, the inlet **202a** of the first working lumen **201a** is located at the proximal end **203** of the main shaft **200**, whereas the inlets **202b** and **202c** of the second and third working lumens **201b** and **201c**, respectively, are located a distance distal to proximal end **203** of the main shaft **200**. According to other implementations one or both of the inlets **201b** and **201c** are located at the proximal end **203** of the main shaft **200**.

(89) In the implementation of FIG. 4A, the working lumens **201a-c** have exit ports located at different longitudinal locations of the main shaft. In the example of FIG. 4C the exit ports are denoted by reference numbers **210**, **211** and **212** with the distal exit port **212**, intermediate exit port **211** and proximal exit port **210** being respectively associated with working lumens **201a**, **201b** and **201c**.

(90) According to some implementations the main shaft **200** also includes a disinfecting fiber lumen **204** wherein which resides a radially emitting fiber **205** that is capable of emitting bacterial disinfecting light. According to some implementations the disinfecting fiber lumen **204** originates at the proximal end **203** of the main shaft **200** and extends distally along at least a portion of the length of the main shaft. The radially emitting fiber **205** enters the disinfecting fiber lumen **204** at the proximal end **203** of the main shaft **200** and runs the entire length or a portion of the length of the disinfecting fiber lumen **204**. Although not required, the disinfecting fiber lumen **204** is preferably placed in between the working lumens **201a-c** as shown in FIG. 4A. In the implementation of FIG. 4A at least a portion of the wall **212** that forms the disinfecting fiber lumen **204** protrudes into each of the lumens **201a-c**. This reduces or eliminates altogether the existence of shadows being cast into the working lumens **201a-c** by virtue of the fiber lumen **204** protruding into the working lumens. The main shaft **200** is a flexible extruded polymer that is transparent or translucent to the light emitted by the radially emitting fiber **205** so that light energy that emanates radially from the radially emitting fiber **205** is able to pass throughout the working lumens **201a-c** and surfaces of the main shaft to disinfect the interior of the working lumens, including the walls that form them, and the exterior surface **209** of the main shaft.

(91) According to some implementations multiple radially emitting fibers **205** are used to disinfect the main shaft **200**. In such instances the fibers may be bundled together inside a common disinfecting fiber lumen or may reside in multiple disinfecting fiber lumens dispersed within the main shaft **200**.

(92) The main shaft **200** may optionally be provided with an imaging lumen **206** that is configured to accommodate another light emitting optical fiber **207**, hereinafter referred to as the “imaging fiber”. As will be discussed in detail below, these elements, in addition to radial openings **208** in the main shaft **200**, form a part of a detection system that may be used to detect biofilm buildup and/or clot formation on the external surface **209** of the main shaft.

(93) According to some implementations of the CVC the main shaft **200** has neither a radially emitting fiber nor an imaging fiber.

(94) According to some implementations the main shaft **200** comprises an elongate proximal end

portion **220** that contains the disinfecting fiber lumen **204** and/or imaging lumen **205**. As will be discussed in more detail below, the elongate proximal end portion **220** facilitates a routing of the radially emitting fiber **205** and/or imaging fiber **207** into the main shaft **200** via the hub **400**. As will be discussed in more detail below, the hub **400** of the CVC is typically encapsulated by a casting material. The end portion **220** serves to prevent the optical fibers from being cast inside the casting material so that the optical fibers maintain a freedom of movement within the hub.

(95) FIGS. **5A** and **5B** illustrate cross-sectional views of a main shaft **200** according to other implementations. In the implementation of FIG. **5A** the disinfecting fiber lumen **204** is centrally located sharing a central axis **215** with main shaft **200** while in the implementation of FIG. **5B** the disinfecting lumen **204** is located apart from the central axis **215** of the main shaft and runs axially parallel with the central axis along at least a portion of the length of the main shaft.

(96) Because the imaging fiber **207** must optically communicate through the radial openings **208** located about the circumference of the main shaft **200**, the imaging lumen **206** is located nearer to the exterior surface **209** of the main shaft than to the central axis **215** of the main shaft as shown in each of FIGS. **4A**, **5A** and **5B**. As stated above, a detailed description of the configuration and function of these elements is provided below.

(97) As explained above, optical fibers typically comprise cylindrical glass or plastic cores through which light is transported. The core runs along the fiber's length and is surrounded by a medium with a lower index of refraction, typically a cladding of a different glass, or plastic. The core and cladding of an optical fiber are susceptible to breaking if excessively stressed. To address this issue, according to some implementations the disinfecting fiber lumen **204** diameter is sized to be larger than the outer diameter of the radially emitting optical fiber **205** so that the optical fiber **205** is capable of sliding inside the disinfecting fiber lumen when the main shaft **200** is bent. This reduces or eliminates altogether the occurrence of tensile stresses in the optical fiber **205**. To this end, according to some implementations the disinfecting fiber lumen **204** has diameter that is up to 30% greater than the outer diameter of the radially emitting fiber **205**.

(98) According to some implementations the main shaft **200** is formed via an extrusion process. In such implementations the radially emitting fiber **205** may be co-extruded with the shaft so that at the end of the extrusion process the fiber **205** resides in the disinfecting fiber lumen **204**. According to a first co-extrusion process the disinfecting fiber lumen **204** is formed to have an inner diameter that is greater than the outer diameter of the radially emitting fiber **205**. This allows the fiber **205** to slide inside the disinfecting fiber lumen **204** as described above. This has an impact of reducing tensile and bending stresses produced within the radially emitting fiber **205** when the main shaft **200** is bent in comparison to stresses that would exist if the fiber **205** were longitudinally fixed in the disinfecting fiber lumen **204**.

(99) Notwithstanding the foregoing, according to a second co-extrusion process the disinfecting fiber lumen **204** is formed to have an inner diameter that is equal to the outer diameter of the radially emitting fiber **205** with the fiber being fixed in the disinfecting fiber lumen.

(100) In use, the CVC is periodically manipulated by a clinician. This manipulation can result in a bending and/or elongation of the main shaft and infusion shaft. In instances when the main shaft and/or infusion shaft contain an optical fiber, this bending and/or elongation of the shaft will induce bending and tensile stresses in the optical fiber. According to some implementations the main shaft **200** is constructed to limit or prevent its bending beyond a minimum bending radius of the radially emitting fiber **205**. (Features for controlling tensile stresses in the optical fiber are discussed below.) The minimum bending radius may be that established by a manufacturer of the fiber **205**. The minimum bending radius may be associated with a function limit or a breaking limit of the optical fiber. A functional minimum bending radius may be specified by the manufacturer of the optical fiber to denote a bending radius of the optical fiber beyond which the optical fiber is unable to properly function. A breakage minimum bending radius may be specified by the manufacturer of the optical fiber to denote a bending radius beyond which a breaking of the core and/or cladding

occurs. Alternatively, the functional minimum bending radius may simply be considered to be an actual bending radius of the radially emitting fiber **205** beyond which the optical fiber is unable to properly function and the breakage minimum bending radius may be considered the actual bending radius of the radially emitting fiber **205** beyond which a breaking of the core and/or cladding occurs. The term “minimum bending radius” as used herein refers to any one of the aforesaid definitions. To achieve a desired minimum bending radius of the main shaft **200** the material used to construct the main shaft may be selected to have a durometer that impedes a bending of the shaft beyond the minimum bending radius. In conjunction with or independent from the material selection, the thickness and geometry of the various walls of the main shaft **200** may be selected to achieve, or assist in achieving, the desired minimum bending radius. Stiffeners (e.g. a coil, mandrel, braiding, etc.) may also be embedded in the main shaft or otherwise attached to the main shaft to achieve the same objective.

(101) According to some implementations the main shaft **200** is: 1) constructed so that the radially emitting fiber **205** is able to slide within the disinfecting lumen **204**, and 2) constructed to limit or prevent the main shafts from bending beyond a minimum bending radius of the radially emitting fiber **205**.

(102) As discussed above, according to some implementations the radially emitting fiber **205**, which includes both the core and cladding, is coextruded with the main shaft **200**. According to other implementations the radially emitting optical fiber **205** is constructed by coextruding a light transmitting core with the main shaft **200** with the shaft material that surrounds the core acting as the fiber cladding. According to such implementations the core abuts the material that surrounds it.

(103) According to some implementations at least a portion of the outer surface **209** of the main shaft **200** is equipped with a reflector that is configured to reduce or impede a loss of light emitted through the main shaft by fiber **205**. This occurs by the reflector reflecting light that would otherwise escape the main shaft back into the main shaft. The reflector may comprise a light reflective coating on the external surface of the main shaft, a light reflective film wrapped about at least a portion of the external surface of the main shaft (e.g. a film heat shrunk on the outer surface of the main shaft), one or more light reflective elements embedded in the walls that form the working lumens **201a-c**, etc.

(104) As illustrated in the figures, according to some implementations the diameter and/or cross-sectional area of the disinfecting radially emitting fiber **205** is smaller than the diameter and/or cross-section area of each of the working lumens **201a-c** of the main shaft **200**. As also shown in the figures, according to some implementations no portion of the working lumens **201a-c** is encircled by fiber **205**.

(105) Catheters in vivo are susceptible to the hazard of blood clot and biofilm formation on the external surface of the catheter. In the case of venous catheters, the blood clot or biofilm may break free from the catheter to create an embolization downstream which can result in serious harm or even death. In the case of urinary catheters, biofilm build up leads to patient discomfort.

(106) To address this issue, according to some implementations the main shaft **200** comprises an ability to detect biofilm and/or blood clot formation on the exterior surface **209** of the shaft by use of an optical based detection system like that illustrated in FIGS. 7A and 7B. The optical based detection system **250** is configured to detect short pulse width back reflections of light **255** reflected off a biofilm and/or blood clot or other unwanted substance residing on the external surface **209** of the main shaft **200**. According to some implementations the detection is accomplished by radially emitting light from the imaging fiber **207** through a plurality of through holes **208** located between the imaging fiber and the exterior surface **209** of the main shaft **200**. According to some implementations the portion of the imaging fiber **207** disposed in the distal section of the main shaft is located near the exterior surface **209** of the main shaft in a spiral fashion as shown in FIG. 6. According to such an implementation the through holes **208** are also spirally dispersed intermittently about the distal section of the main shaft **200**, as also shown in

FIG. 6, with each through hole **208** being longitudinally spaced from an adjacent through hole. Light delivered into the through holes **208** is reflected back into the through holes **208** and delivered to a back reflectance detector **252** as shown in FIG. 7B. The arrival time and amplitude of the light reflected back will depend on the location, density, thickness and mass of the biological tissue from which it is reflected. According to some implementations the imaging fiber **207** is fixed inside the main shaft **200**.

(107) Pulsed light **254** is delivered to the imaging fiber **207** from a light source **251** through a beam splitter **253**. The light source **251** is typically a laser. The same beam splitter **253** is used to direct the back reflected light pulses **255** to the back reflectance detector **252**. The pulse width of the light and the longitudinal distance between the through holes **208** in the main shaft **200** are selected such that the pulse width is shorter than the time taken by the light to travel between the through holes. As a result, pulsed back reflective light **255** can be consecutively collected through the through holes **208**, beginning with the proximal-most through hole **208** and ending with the distal-most through hole **208**, and delivered to the back reflectance detector **252** without colliding with one another. The back reflectance detector **252** identifies the through hole from which the back reflected light pulses originate by the order in which it receives the back reflected light pulses. The back reflectance detector **252** converts the back reflected light pulses **255** into electrical signals that are quantified into digital data that may be processed by a computer processor. According to some implementations a computer processor is provided in the back reflectance detector itself, while in other implementations the processor is located outside the back reflectance detector.

(108) Upon the train of back reflected light pulses **255** having been received by the back reflectance detector **252**, data associated with these pulses is processed by a software algorithm implemented by the processor to determine whether or not a biofilm, clot or other unwanted substance has formed on the exterior surface **209** of the main shaft **200**. This can be achieved by comparing a baseline set of data representative of a clean outer exterior surface **209** of the main shaft **200** with the data collected. If the collected data differs from the baseline set of data in a particular way, the software algorithm may determine that a biofilm and/or clot exists on the outer surface of the main shaft.

(109) According to some implementations the data is stored in a computer readable memory and the software algorithm compares the collected data collected with the stored data to determine if a biofilm or clot exists.

(110) According to other implementations, and/or in conjunction with one or more of the processing methods described above, a software algorithm may process the collected data to form an image that can be used to determine the existence of a biofilm and/or clot on the outer surface **209** of the main shaft **200**.

(111) In accordance with any of the comparison methods disclosed above, a control unit **259** of the detection system may be configured to automatically activate the light source **251** to initiate a disinfection of the main shaft **200** upon the control unit determining or being informed of an unwanted substance existing on the outer surface of **209** of the main shaft **200**.

(112) According to some implementations the imaging fiber **207** is an optical fiber in which selected portions of the cladding have been removed at the site of the through holes **208** in the main shaft **200**. Light is delivered from the fiber into and through the through holes **208** of the main shaft **200** via the holes formed in the cladding. According to other implementations the imaging fiber **207** comprises a radially emitting fiber **205** like those described above. According to such implementations the radially emitting fiber may possess an outer most layer (e.g. a jacket) having a plurality of strategically placed apertures that are circumferentially disposed about the radially emitting fiber **207**. Light is delivered from the fiber **207** into and through the holes **208** in the main shaft **200** via the apertures formed in the outer most layer.

(113) An advantage of the optical based detection systems disclosed herein is that they do not require a mechanical rotation and translation of the imaging fiber **207** to obtain three-dimensional

data pertaining to the outer surface condition of the main shaft.

(114) The introduction of fluids or treatment instruments into the working lumen or lumens of a CVC occurs via the use of one or more infusion shafts that are in fluid communication with the one or more working lumens of the main shaft. As discussed above, in the example CVC **100** of FIG. **3** there are three infusion shafts **300** that each have a working lumen **308** that is respectively in fluid communication with the first, second and third working lumens **201a**, **201b**, **201c** of the main shaft **200**. According to some implementations communication between the working lumens **308** of the infusion shafts **300** and the working lumens of the main shaft **200** occurs via conduits formed in the hub **400**. A detailed description of various implementations of the hub **400** is provided below.

(115) FIG. **8A** illustrates an external perspective view of an infusion shaft **300** according to one implementation. FIG. **8A** also depicts an infusion clamp **340** disposed on the infusion shaft **300** that is used to clamp down on the tubular body **303** of the infusion shaft **300** to occlude flow through the infusion shaft. A more detailed description of the infusion clamp **340** is provided below.

(116) FIG. **8B** is an exploded view of the infusion shaft **300** of FIG. **8A** (without the infusion clamp **340**) that shows a partial view of a radially emitting fiber **301** and an end emitting fiber **302** that extend along at least a portion of the length of the infusion shaft. As will be discussed in more detail below, the radially emitting fiber **301** and end emitting fiber **302** are used to deliver disinfecting light, such as blue light, to the infusion shaft **300** and/or proximal connector **304**.

(117) According to some implementations the infusion shaft **300** possesses only one of a radially emitting fiber **301** and an end emitting fiber **302**. According to other implementations the infusion shaft possesses neither a radially emitting fiber nor an end emitting fiber.

(118) In the example of FIGS. **8A** and **8B** the infusion shaft **300** includes the tubular body **303** that is connected at its proximal end to a connector **304**. According to some implementations radially extending barbs **305** are provided on the distal end portion of the connector **304** to facilitate an attachment of the tubular body **303** to the connector **304**. A removable cap **306** is connected to a proximal end of the connector **304** via a threaded connection. As used herein, the cap **306** is considered to be a part of the proximal connector **304**. A strain relief element **307** may also be provided at the junction of the tubular body **303** and the connector **304**.

(119) FIG. **9A** is a cross-section view of the tubular body **303** according to one implementation. The tubular body **303** comprises an extruded polymeric structure having walls that define a working lumen **308**, a radially emitting fiber lumen **309** configured to house the radially emitting fiber **301**, and an end emitting fiber lumen **310** configured to house the end emitting fiber **302**. According to some implementations the radially emitting fiber lumen **309** and end emitting fiber lumen **310** are at least partially located in a key portion **311** of the polymeric structure of the infusion shaft **300** with the radially emitting fiber lumen **309** being located adjacent the working lumen **308**. According to some implementations the radially emitting fiber lumen **309** is defined by a wall segment **312** that protrudes into the working lumen **308** as shown in FIG. **9A**.

(120) According to such an implementation the radially emitting fiber lumen **309** may, but not necessarily, overlap with a portion of the working lumen **308** for the purpose of maximizing the dispersion of the disinfecting light inside the working lumen when the radially emitting fiber **301** is illuminated. This maximization occurs, at least in part, as the result of no shadows being cast into the working lumen **308** by virtue of the wall segment **312** protruding into the working lumen **308** and/or an overlapping of fiber lumen **309** with the working lumen **308**.

(121) As illustrated in the figures, according to some implementations the diameter and/or cross-sectional area of each of the radially emitting fiber **301** and end emitting fiber **302** is smaller than the diameter and/or cross-section area of the working lumens **308** of the infusion shaft **300**. As also shown in the figures, according to some implementations no portion of the working lumen **308** is encircled by either of fibers **301** and **302**.

(122) As explained above, the core and cladding of an optical fiber are susceptible to breaking if excessively stressed. To address this issue, according to some implementations the diameter of the

radially emitting fiber lumen **309** is sized to be larger than the outer diameter of the radially emitting optical fiber **301** so that the optical fiber is capable of sliding inside the radially emitting fiber lumen **309** when the infusion shaft **300** is bent. To this end, according to some implementations the radially emitting fiber lumen **309** has diameter that is up to 30% greater than the outer diameter of the radially emitting fiber **301**.

(123) To facilitate a closing off of the working lumen **308** by use of the clamp **340**, portions **313** of the sidewalls that form the working lumen have a reduced thickness t_1 as compared to the remaining portions of walls that define the working lumen **308**. The reduced thickness portions **313** are engineered to act as hinges that promote a collapsing of the tubular body **303** on itself when the clamp **340** is caused to exert a force F against a compression wall **314** that is located opposite the key portion **311**. According to some implementations the compression wall **314** includes a recess **315** that is configured to mate with the protruding wall segment **312**. The mating of the parts **312** and **315** promotes an ordered closure of the working lumen **308** in a manner that assists in inhibiting a twisting of the tubular body **303** when the clamp **340** acts to close the working lumen.

(124) The key portion **311** of the infusion shaft **300** serves a number of functions. It provides a region that is located outside, or substantially outside, the working lumen **308** for housing the radially and end emitting fibers **301**, **302**. This is important because if the fibers **301**, **302** were to be located outside the key portion **311**, they would be susceptible to breakage when the tubular body **303** is clamped to effectuate a closing of the working lumen **308**. A further advantage of housing the fibers **301**, **302** in the key portion **311** of the tubular body **303** is that accessibility to and through the working lumen **308** of the infusion shaft **300** remains the same or similar to that of conventional CVC systems.

(125) As will be described in more detail below, the key portion **311** can also be configured to reside within a groove **346** of a lower pad **342** of the clamp **340** in order to maintain the tubular body **303** properly oriented in the clamp to ensure the upper pad **341** of the clamp properly acts on the compression wall **314** during the clamping procedure. This prevents or inhibits a twisting of the tubular body **303** during the clamping procedure that could otherwise result in a breaking of the fibers **301**, **302**.

(126) As explained above, stresses produced within an optical fiber (e.g., a radially emitting optical fiber and/or an end emitting optical fiber) can be caused by a bending of the fiber. For this reason, according to some implementations the infusion shaft **300** is constructed to limit or prevent its bending beyond a minimum bending radius of one or both of the radially emitting fiber **301** and end emitting fiber **302**.

(127) As discussed above in conjunction with the description of the main shaft **200**, optical fibers typically have a functional minimum bending radius beyond which the fibers are unable to properly function and a breakage minimal bending radius beyond which a breaking of the core and/or cladding occurs. The functional and/or breakage minimum bending radius may be specified by the manufacturer of the optical fiber. The functional and/or breakage minimum bending radius may alternatively be considered the actual bending radius of the optical fiber beyond which the optical fiber is unable to properly function or which a breaking of the core and/or cladding occurs. To achieve a desired minimum bending radius of the infusion shaft **300** the material used to construct the infusion shaft may be selected to have a durometer that impedes a bending of the shaft beyond the functional or breakage minimum bending radius of one or both of the radially emitting fiber **301** and the end emitting fiber **302**. In conjunction with or independent from the material selection, the thickness and geometry of the various walls of the infusion shaft **300** may be selected to achieve, or assist in achieving, the desired function or breakage minimum bending radius. Moreover, a stiffening of the infusion shaft **300** may occur by embedding one or more stiffening elements (e.g. a coil, mandrel, braiding, etc.) inside the key portion **311** along a length of the tubular body **303**. According to other implementations a stiffening jacket or other element may be attached to at least a portion of the outer surface of the tubular body to obtain a desired stiffness

that resists against the fiber **301** and/or **302** bending beyond its/their minimum bending radius. Stiffeners (e.g. a coil, mandrel, braiding, etc.) may also be embedded in the main shaft or otherwise attached to the main shaft to achieve the same objective.

(128) Because a majority of the walls of the tubular body **303** that form the working lumen **308** are engineered to be relatively flexible to accommodate a closing of the working lumen by the clamp **340**, according to some implementations a majority of the stiffness attributed to the tubular body **303** is engineered to reside in the key portion **311**. According to some implementations this is accomplished by the key portion **311** having an overall width dimension “w” that is greater than any of the thickness dimensions of the remainder of the walls that form the working lumen **308**. The shape of the key portion **311**, for example a rectangular shape, can also contribute to its stiffness.

(129) With reference to FIGS. **9A** and **9B**, according to one implementation the dimensions of the tubular body **303** are as follows. The overall height “h.sub.1” is 3.055 millimeters with a horizontal centerline of the working lumen **308** residing 1.850 millimeters above the bottom end of the tubular body as denoted by the reference “h.sub.2”. The key portion **311** has a height “h.sub.3” and a width “w”, each of which are 1.000 millimeters. The centerline of the end emitting fiber lumen **310** is located a height “h.sub.4” of 0.425 millimeters above the bottom end of the tubular body **303**. Each of the walls of the tubular body that form the hinge portions **313** have a thickness “t.sub.1” of millimeters. The wall thickness dimensions “t.sub.2” and “t.sub.3” are 0.250 millimeters and 0.100 millimeters, respectively. The distance “d” between the central axis of the radially emitting fiber lumen **309** and the horizontal centerline of the working lumen **308** is 0.850 millimeters. The diameter of each of the radially emitting fiber lumen **309** and end emitting fiber lumen **310** is about millimeters. Furthermore, the width “w” of the key portion **311** is dimensioned to be smaller than the maximum width dimension of the working lumen **308**.

(130) According to some implementations the ratio of h.sub.3/h.sub.1 is less than 0.6. By limiting the size of the key portion **311** in this way, it limits the degree by which the infusion shaft **300** differs from traditional infusion shafts. In addition, according to some implementations the ratio of t.sub.1/t.sub.2 is less than 1.0 to yield the creation of hinge portions **313** that are configured to bend when a force is applied downward on the top end of the infusion shaft **300**.

(131) FIGS. **10A-C** are cross-section views of example alternative infusion shaft tubular body designs. In the implementation of FIG. **10A** the working lumen **308** of the tubular body **303** has a semi-circular configuration that is partially truncated by a structure **320** that houses the radially emitting fiber lumen **309**. The structure **320** extends from the key portion **311** that houses the end emitting fiber lumen **310**. Clamping the working lumen **308** shut in the direction of arrow A can occur in a manner like that described above in conjunction with the implementation of FIG. **9A**.

(132) In the implementations of FIGS. **10B** and **10C** each of the radially emitting fiber lumens **309** and end emitting fiber lumen **310** are suspended in the tubular body **303** by a septum/wall **321** with the radially emitting fiber lumen residing centered in the tubular body. In each of these implementations, the infusion shaft is provided with first and second lumens **308a** and **308b** that are separated by the septum **320**. In each of these implementations the first and second working lumens **308a** and **308b** can be clamped shut by applying an inward force F1 and F2 to the walls of the tubular body, respectively. In each of the implementations of FIGS. **10A** and **10B** first and second wings **321a** and **321b** protrude from opposite sides of the tubular body and are used to align the tubular body **303** in a clamp (not shown) that is capable of applying the forces F1 and F2. According to some implementations the working lumen **308a** is configured to transport a fluid and the working lumen **308b** is configured to receive a treatment instrument, such as, for example, a guidewire.

(133) As shown in FIG. **10B**, the radial emitting fiber lumen **309** is partially formed of wall segments on each side of the fiber lumen that protrude into the respective first and second lumens **308a** and **308b**. Although not shown in FIG. **10B**, according to some implementations the interior

wall **322** of the working lumens **308a** and **308b** each comprise a recess that is configured to mate with the protruding wall segments of the fiber lumen. The mating of the parts promotes an ordered closure of the working lumens **308a** and **308b** in a manner that assists in inhibiting a twisting of the tubular body **303** when the clamp **340** acts to close the working lumens.

(134) According to some implementations the infusion shaft **300** is a flexible extruded polymer that is transparent or translucent to the light emitted by the radially emitting fiber **301** so that light that emanates radially from the radially emitting fiber **301** is able to pass through the walls of the infusion shaft to disinfect the interior walls **322** of the working lumens **308a**, **308b**.

(135) As stated above, according to some implementations the infusion shaft **300** is formed via an extrusion process. In such implementations one or both of the radially emitting fiber **301** and end emitting fiber **302** may be co-extruded with the infusion shaft so that at the end of the extrusion process fiber **301** resides in the radially emitting fiber lumen **309** and fiber **302** resides in the end emitting fiber lumen **310**. According to a first co-extrusion process one or both of the radially emitting fiber lumen **309** and end emitting fiber lumen **310** is formed to have an inner diameter that is greater than the outer diameter of the radially emitting fiber **301** and end emitting fiber **302**, respectively. This allows the fibers **301** and **302** to slide inside their respective lumens. This has an impact of reducing tensile and bending stresses produced within the fibers when the infusion shaft **300** is bent in comparison to stresses that would otherwise exist if the fibers were longitudinally fixed inside their lumens. According to some implementations one or both of the radially emitting fiber lumen **309** and end emitting fiber lumen **310** has diameter that is up to 30% greater than the respective outer diameter of the radially emitting fiber **301** and end emitting fiber **302**.

(136) According to some implementations the infusion shaft **300** is: 1) constructed so that one or both of the radially emitting fiber **301** and end emitting fiber **302** is able to slide within its respective lumen **309** and **310**, and 2) constructed to limit or prevent the infusion shaft from bending beyond a minimum bending radius associated with one of the radially emitting fiber **301** and end emitting fiber **302**.

(137) FIG. **11A** shows an implementation of the clamp **340** used to close off flow through the infusion shaft **300** with the clamp being in the open position. FIG. **11C** illustrates the clamp of FIG. **11A** being in the closed position. FIG. **11B** illustrates the infusion shaft **300** running through the clamp of FIG. **11A** with the clamp being in the open position.

(138) Because the core and cladding of optical fibers **301** and **302** are susceptible to breakage when the infusion shaft is bent and/or twisted, according to some implementations the clamp **340** is keyed at its proximal and distal ends to the infusion shaft **300** as shown in FIG. **11B** to prevent the shaft from twisting during a closing of the clamp onto the shaft. As discussed above, the groove **346** in the lower pad **342** serves the same function. To further guard against the breakage of the optical fibers **301** and **302** during and after the closure of the clamp **340**, according to some implementations the upper and lower pads **341** and **342** of the clamp are compliant enabling them to at least partially conform to the outer profile of infusion shaft. This has the effect of distributing the force applied by the clamp **340** over a larger surface area of the infusion shaft **300**. According to some implementations the upper pad **341** is assembled on a resilient arm **344** and is moved to the closing position by the application of a downward force at an end of the arm **344**. According to some implementations the clamp **340** is equipped with a stop that engages the upper pad **341** and/or arm **344**. The clamp **340** is configured with the stop to limit the amount by which the upper pad **341** may be moved downward against the tubular body **303** of the infusion shaft **300**. The downward movement of the upper pad **341** is restrained sufficiently to limit the amount of force it is capable of applying to the tubular body **303** so as to inhibit or prevent a breaking of one or both of fibers **301** and **302** while at the same time accommodating a closing of the infusion shaft lumen **308**. As shown in FIG. **11A**, according to one implementation the stop **354** is in the form of one or more tabs that protrudes outward from the distal wall **353** of the clamp. The tabs **354a** are configured to abut bottom portions **354b** of the arm latch **355** to limit downward movement of the

arm **344** and upper pad **341**.

(139) According to some implementations the clamp **340** includes a base **343** and an arm **344** that are connected by a vertically extending proximal wall **345**. A hinge **347** provided between the proximal wall **345** and arm **344** enables the arm to be resiliently moved downward toward the base **343** when a force is applied to the top surface of the arm. Slip resistant grips **348** and **349** are respectively provided on both the top surface of the arm **344** and on the bottom surface of the base **343**. According to some implementations the grips **348** and upper pad **341** comprise a unitary structure with the grips protruding upward from the upper pad into apertures formed in the arm **344**. Likewise, according to some implementations the grips **349** and lower pad **342** comprise a unitary structure with the grips protruding downward from the lower pad into apertures formed in the base **343**.

(140) Keying the clamp **340** to the infusion shaft **300** occurs by a passing of the infusion shaft **300** through key openings **350** and **351** in the clamp having a same or similar profile of the shaft. In the implementation of FIGS. **11A-C** key opening **350** extends through the proximal wall **345** and through a proximally extending protrusion **352**. Although the proximally extending protrusion **352** is not required, it advantageously contributes to preventing a twisting of the infusion shaft in the clamp **340** by increasing the length of the infusion shaft **300** that is restrained. Key opening **351** is located in a distal wall **353** of the clamp.

(141) A closing of the clamp **340** from an open position to a closed position occurs when a downward force is applied to the top surface of arm **344**. A distal end of the arm **344** includes a latch **355** having an upward facing shelf **356** that is configured to engage with a downward facing shelf **357** on the distal wall **353** to lock the clamp in a closed position as shown in FIG. **11C**. During the closing operation the latch **355** moves downward along the distal wall **353** over an outward sloping ramp **359** until the upward facing shelf **356** surpasses the bottom of the ramp and moves inward to rest against the downward facing shelf **357**. As the latch **355** of the arm **344** passes along the ramp **359**, the distal wall **353** of the clamp flexes slightly outward and then attempts to return to its original position when the upward facing shelf **356** surpasses the bottom of the ramp **359**. The resilient nature and configuration of the arm **344** and distal wall **353** cause the shelves **356** and **357** to be pressed against one another to maintain the clamp **340** in a closed position as shown in FIG. **11C**.

(142) To return the clamp **340** to its open position a rearward force is applied to the curved latch release **360** that extends upward from the distal wall **353** to flex the distal wall in the direction A as shown in FIG. **11C**. This has an effect of sliding the downward facing shelf **357** of the distal wall **353** off the upward facing shelf **356** of latch **355**. When this occurs, the arm **44** returns to its biased open position as shown in FIGS. **11A** and **11B**.

(143) According to some implementations the clamp **340** and the proximal connector **304** of the infusion shaft **300** comprise a unitary structure with the clamp protruding from the distal end of the proximal connector. According to such implementations the proximal connector **304** and clamp **340** may be injection molded as a single piece. The proximal connector is discussed in detail below.

(144) As discussed above, the working lumen **308** of the infusion shaft **300** is coupled to a working lumen of the main shaft **200** through a connection that occurs inside the hub **400**. A detailed description of the hub **400** is provided below. In addition, according to some implementations the radially emitting fiber **301** and end emitting fiber **302** are introduced into the key portion **311** of the infusion shaft **300** via the hub **400**. Thus, according to some implementations both the radially emitting fiber **301** and end emitting fiber **302** run the entire length of the tubular body portion **303** of the infusion shaft **300** as shown in FIG. **12**, while in other implementations the radially emitting fiber **301** runs less than the entire length of the tubular body. As discussed above, the tubular body **303** is connected to a proximal connector **304** through which therapeutic agents and/or medical instruments are introduced into the infusion shaft **300**. FIG. **13** shows a perspective view of a proximal end segment **319** of the tubular body **303** according to one implementation.

(145) FIG. 14A illustrates a cross-section side view of a proximal connector 304 to which the proximal end of the tubular body 303 of FIG. 12 is attached. FIG. 16 is an external perspective view of the proximal connector of FIG. 14A. FIG. 14B shows a partial cross-section view of the proximal connector 304 with the end segment 319 of the key portion 311 of the tubular body 303 fully inserted into a receiving lumen 363 of the proximal connector. According to some implementations the entirety of proximal connector is made of a material that is transparent or translucent to the light emitted by the radially emitting fiber 301. According to other implementations only selected portions of the proximal connector 304 located adjacent radially emitting fiber 301 is made of a material that is transparent or translucent to the light emitted by fiber 301. In either case, the contents and/or walls of at least a portion of the inner lumen 365 of the proximal connector 304 can be exposed to the disinfecting light emitted by the radially emitting fiber 301.

(146) In the implementation of FIG. 12 the tubular body 303 comprises the working lumen 308 and the radially emitting and end emitting fibers 301, 302 residing in their respective lumens 309, 310 inside the key portion 311 of the tubular body. A proximal end section of the tubular body 303 is skived so that the end segment 319 of the key portion 311 extends proximal to the proximal end of the working lumen 308. The distal end 323 of the tubular body 303 that resides inside the hub 400 may also be skived. As shown in FIGS. 12, 14A, and 14B, an end portion 301a of the radially emitting fiber 301 and an end portion 302a of the end emitting fiber 302 protrude from the proximal end of the key portion 311 of the tubular body 303 and reside respectively within lumens 361 and 362 located in the proximal connector 304. In the implementation of FIG. 14B the end portions 301a and 302a of the fibers 301 and 302 are shorter than the respective lumens 361 and 362 they reside in. Although the length of the fibers 301 and 302 can be selected to cause their ends 301c and 302c to abut the end walls 361a and 362a of the lumens 361 and 362, the provision of gaps as shown in FIG. 14B advantageously allows for less restrictive tolerances (reducing manufacturing costs) and minimizes the risk of damaging the fibers 301 and 302 when the CVC 100 is assembled.

(147) Again with reference to FIG. 14A, the proximal connector 304 further comprises features for directing the light 366 emitted from the end emitting fiber 302 into the internal space of the proximal connector. FIG. 15 shows a ray tracing of the light 366 dispersed inside the proximal connector 304 according to one implementation. The light emitted by the end emitting fiber 302 passes through a first optical surface/lens 367 and a second optical surface/lens 368 that act to alter the trajectory of the light so that it impinges on a light reflector 369 positioned on, or otherwise integrated into, a distal facing surface of a central portion 306a of the cap 306. According to some implementations the first and second optical surfaces 367 and 368 have an RMS surface roughness of less than 180 angstroms.

(148) According to some implementations the trajectory of the disinfecting light is altered as a result of being refracted by each of the first and second optical surfaces 367 and 368. Optical surfaces of this type are also referred to herein as “refractive optical surfaces”. (Refraction is a deflection from a straight path undergone by a light ray or energy wave in passing obliquely from one medium (such as air) into another (such as glass) in which its velocity is different.) That portion of the proximal connector 304 residing between the first and second optical surfaces is a material that is substantially transparent or at least translucent to the light emitted by the end emitting fiber 302. According to some implementations the material is a Teflon or a polycarbonate.

(149) In the implementation of FIG. 15, the first and second optical surfaces/lenses 367, 368 and the light reflector 369 are arranged in the proximal connector 304 to cause the light emitted from the end emitting fiber 302 to flood at least a portion of the length of the inner lumen 365 of the proximal connector and/or at least a portion of the inner lumen 308 of the tubular body 303. According to some implementations this length of the inner lumen 365 is out of reach of the disinfecting light emitted by the radially emitting fiber 301. In the implementation of FIG. 15,

however, there is an overlapping of the light paths that emanate from fibers **301** and **302** along at least a portion of the length of the inner lumen **365**. Thus, when the fibers **301** and **302** are both illuminated this has an effect of providing not just one, but two doses of disinfecting light in the overlapping region.

(150) According to some implementations an index matching material, such as a gel or adhesive, is positioned in the gap that separates the end **302c** of the end emitting fiber **302** from the end wall **362a** of lumen **362**. The index matching material is selected to have a refractive index between that of the core of the end emitting fiber **302** and that of the first optical surface **367** formed in or located on the end wall **362a** of lumen **362**. During an assembly of the CVC **100** the index matching material may be introduced into the gap via a port **370** located in a side of the proximal connector.

(151) The first optical surface/lens **367** resides at the inner surface of the end wall **362a** of lumen **362**. According to some implementations the inner surface of the end wall **362a** is of optical quality that minimizes a scattering of the light emitted by fiber **302** so that the vast majority of the disinfecting light entering the surface/lens **367** is directed onto the second optical surface/lens **368**. In a like manner, the second optical surface/lens **368** of optical quality so that a vast majority of the disinfecting light that enters the surface/lens **368** is transported to the cap light reflector **369** and then onto the contents and/or inner walls of the inner lumen **365** of the proximal connector.

(152) According to some implementations the first optic surface/lens **367** is arranged perpendicular to a longitudinal axis **373** of the proximal connector **304** and the second optical surface/lens **368** is positioned at an angle relative to the longitudinal axis. The angle by which the second optical surface/lens **368** is tilted with respect to the longitudinal axis **391** is selected to assist in diverting the light exiting the lens upward toward a selected portion of the cap reflector **369**. According to some implementations the lens **368** is tilted an angle less than 20 degrees in relation to the plane orthogonal to the longitudinal axis **391**. According to some implementations, like that shown in FIG. **19A**, the second optical surface/lens **368** is dispensed with and the first optical surface/lens **367** is angled to assist in directing the light emitted by the end emitting fiber **302** to impinge on a location of the cap light reflector **369** that causes the light to flood at least a portion of the length of the internal lumen **365** of the proximal connector and/or at least a portion of the internal lumen **308** in the tubular body **303**. An exemplary light path of such implementations is shown by the arrows in FIG. **19A**. This can occur by bending the distal end portion of the lumen **362** upward so that it is not parallel to the longitudinal axis **391** of the proximal connector **304** as shown in FIG. **19A**. In this way a direct optical path between the first optical surface/lens **367** and the reflector **369** may be established.

(153) The cap light reflector **369** may comprise any shape that is capable of causing the light impinged upon it to be reflected down at least a portion of the inner lumen **365** of the proximal connector **304**. According to some implementations the cap reflector **369** comprises a concave or convex surface having a radius of curvature suitable for establishing a desired light path down the inner lumen **365** of the proximal connector. According to other implementations the reflector **369** may comprise a flat surface that may or may not be oriented at an angle in relation to the longitudinal axis **373** of the proximal connector **304**. FIG. **18** is a perspective view of the proximal connector cap **306** according to one implementation. As also shown in FIG. **14A**, the cap **306** has internal threads **371** that mate with external threads **372** of the proximal connector that allow that cap to be fixed to and removed from the proximal connector.

(154) As best seen in FIGS. **14A** and **17**, according to some implementations the proximal connector **304** is equipped with a ramp **373** that is configured to facilitate an advancement of a medical instrument through the lumen **365** of the proximal connector. The medical instrument may be, for example, a guidewire. In the implementation of FIG. **17** the ramp **373** comprises a first part **373a** and a second part **373b** that reside on opposite sides of the second optical surface **368** so that there exist a direct line of sight between the second optical surface/lens **368** and the cap reflector

369.

(155) In accordance with some implementations disclosed herein each of the radially emitting fiber **301** and end emitting fiber **302** emits blue light at a wavelength of between 380-495 nm and an exposure of between 100-1,000 Joules/cm² is employed to kill the unwanted bacteria. According to other implementations fibers **301** and **302** emit ultra-violet light at a wavelength of 100-400 nm and exposure up to 6 J/cm² to kill unwanted bacteria.

(156) In the infusion shaft **300** implementations discussed above, both a radially emitting fiber **301** and an end emitting fiber **302** are used to effectuate a disinfection of the infusion shaft. However, according to other implementations only one of a radially emitting fiber and an end emitting fiber is used. For example, according to some implementations disinfection predominately occurs inside the proximal connector **304** by use of the end emitting fiber **302** and the radially emitting fiber **301** is dispensed with altogether as shown in FIG. **19A**. According to other implementations the end emitting fiber **302** is dispensed with and the radially emitting fiber **301** alone is used to disinfect at least a portion of the tubular body **303** and at least a portion of the proximal connector **304** of the infusion shaft **300** as shown in FIG. **19B**. According to implementations that do not include an end emitting fiber the radially emitting fiber **302** may extend proximally inside the proximal connector **304** to a location near or adjacent the distal end of the cap **306** when the cap is fully threaded onto the connector as shown in FIG. **19B**. In this manner the radially emitting fiber **301** is capable of disinfecting a greater length of the proximal connector.

(157) According to some implementations the radially emitting fiber **301** is also configured to emit light from an end thereof. According to such implementations the light emitted from the end of the radially emitting fiber **301** may be delivered to a target location inside the proximal connector **304** and/or a target location inside the tubular body **303** of the infusion shaft by the use of one or more optical surfaces and/or one or more reflectors similar to the concepts described above and below in conjunction with the examples of FIGS. **14A**, **14B**, **20A**, **20B**, **21A** and **21B**. For example, as shown in FIG. **19C**, disinfecting the infusion shaft **300** may occur by use of a radially emitting fiber **301** that emits light radially along its length and also emits light from a distal end of the fiber. Like the implementation of FIG. **19A**, an optical surface **367** and reflector **369** may be used to distribute the light emitted from the end of the radially emitting fiber **301** into the proximal connector.

(158) A majority of the light transmitted by a radially emitting fiber is typically emitted radially from the fiber with a smaller amount being emitted from the end of the fiber. For example, the Fibrance® Light Diffusing Fiber manufactured by Corning® Incorporated has a diffusion length that is characterized by an emitting of 90% of its light radially along its length with the remaining 10% being emitted from its end. These percentages will vary among different radially emitting fibers and are typically established by the fiber manufacturer. Because it may be desirable to use the light emitted from the end of the radially emitting fiber to disinfect an object, like in the example of FIG. **19C**, the percentage of light emitted through the end of the radially emitting fiber may be increased by cutting short the length of the fiber. For instance, a radially emitting fiber that has a given diffusion length that results in 90% of its light being radially emitted with the remaining 10% being emitted from its end can be cut short to cause a percentage increase in the amount of light being emitted from the end. For example, the radially emitting fiber may be cut by a certain length to cause a decrease in the amount of light that is radially emitted and an increase in the amount of light that is end emitted. According to some implementations the radially emitting fiber is cut short to cause 50-80% of the light to be radially emitted and correspondingly of the light to be end emitted.

(159) According to other implementations the radially emitting fiber may itself be engineered to obtain a desired distribution of light from its sides and end. This can be achieved, for example, by altering the amount of nano-sized structures doped within the fiber. According to some implementations the radially emitting fiber is engineered to cause 50-90% of the light to be radially emitted and correspondingly 50-10% of the light to be end emitted. According to other

implementations the radially emitting fiber is engineered to cause 50-80% of the light to be radially emitted and correspondingly 50-20% of the light to be end emitted.

(160) It is important to note that multiple radially emitting fibers and/or multiple end emitting fibers may be used to disinfect the infusion shaft **300**.

(161) In regard to both the main shaft **200** and the infusion shaft **300**, when a radially emitting fiber that emits light from an end thereof is used, the end of the radially emitting fiber may be coated with a reflective material, such as gold, to direct the light that would otherwise be lost out the end of the fiber back into the fiber core. Alternatively, the end of the radially emitting fiber may be fitted within a cap that has an internal reflective surface that causes at least a portion of the light emitted from the end of the fiber back into the fiber. Reflecting the light back into the fiber advantageously increases the amount of light emitted along the length of the radially emitting fiber.

(162) The laser induced damage threshold, or the ability to propagate light of a given power density, varies among material types. For example, a fiber optic core comprised of fused silica glass may have a higher tolerance to laser damage than a polymer. The power density of light emitted by a multimode type end emitting fiber can be high as a result of essentially the totality of the light input being emitted from the end of the small diameter core. The power densities of the light emitted by the multimode type end emitting fiber may exceed the capabilities of the materials selected for injection molded optics or casted hubs. To alleviate this problem, according to some implementations the end emitting fiber **302** is equipped with an end cap that reduces the power density of the light that leaves the end emitting fiber core.

(163) FIG. **48A** shows a perspective side view of an end portion of an end emitting fiber **650** that has an end cap **651** attached to the end of the fiber **650**. FIG. **48B** shows a cross-section of the end cap according to one implementation. The fiber includes a glass or polymer core **652** and a cladding **653** disposed about the core. The end cap **651** includes a cylindrical body **654** that houses a medium **655** that is optically coupled to the fiber core **652**. The medium is transparent to the light it receives from the core **652** and has an index of refraction that is the same or close to that of the fiber core **652**. The medium is sized and shaped so that the light that exits the end **656** of the end cap **651** has a power density that is lower than the power density of the light when it leaves the fiber core **652**. This is accomplished by widening the light beam received from the fiber core. The widening of the light beam occurs as a result of the medium having a greater diameter than that of the fiber core. In the example of FIG. **48B**, the diameter of the medium **655** increases between its proximal and distal ends **657** and **656**, respectively. However, according to other implementations the medium may be cylindrical or spherical in shape. Coupling of the end cap medium **655** to the fiber core **652** may occur in several ways. According to one method the proximal end of the medium is fused with the fiber core. According to other implementations the medium is formed from the fiber core.

(164) FIG. **9C** illustrates a cross-section of the tubular body **303** of an infusion shaft whereby a curved reflector of light **316a** is embedded partially surrounding the radially emitting fiber **301** to direct light emitted from the sides and bottom of the fiber **301** upward toward the internal lumen **308**. FIG. **9D** illustrates a similar configuration wherein the reflector **316b** comprises a flat profile extending across at least a portion of the width of the key portion **311** of the tubular body **303**. In implementations that also involve the use of an end emitting fiber **302**, the reflector **316** may be located between the radially emitting fiber lumen **309** and the end emitting fiber lumen **310**. According to some implementations the reflectors **316a** and **316b** are embedded into the tubular body **303** of the infusion shaft **300** while the tubular body is being extruded via co-extrusion process.

(165) According to other implementations, that may or may not include a light reflector like that of reflector **316a** or reflector **316b**, at least a portion or all of the peripheral wall of the tubular body **303** is coated, wrapped or impregnated with a material that is light reflective. This advantageously results in light being reflected back into the tubular body **303** (that would otherwise escape),

thereby increasing the light density within the tubular body when the radially emitting fiber is illuminated. In the implementation of FIG. 9E an outer thickness of the peripheral wall 317 of the tubular body 303 is impregnated with light reflective materials, such as, for example, aluminum particles. According to some implementations the light reflective material is impregnated into the peripheral wall during the extrusion of the tubular body 303. In the implementation of FIG. 9F the outer surface of the tubular body 303 is coated or wrapped with a light reflective material 318. According to implementations in which the outer surface of the tubular body 303 is wrapped, the wrapping may comprise a heat shrink having light reflective properties that is heat shrunk onto the outer surface. The outer surface of the tubular body 303 may also comprise a reflective film, such as a layer of light reflective paint.

(166) According to some implementations at least a portion of the external surface of the proximal connector 304 is coated or wrapped with a light reflective material so that at least a portion of the disinfecting light delivered into the internal lumen 365 is reflected back into the lumen when the light impinges on the light reflective material. Like the tubular body 303, the proximal connector 304 may be tightly wrapped with a heat shrink that possesses a light reflective material. The external surface of the proximal connector 304 may also comprise a reflective film, such as a layer of light reflective paint. The use of the light reflective material improves internal disinfection and acts to reduce or eliminate an impingement of the disinfecting light onto the skin of the patient that could otherwise result in patient discomfort in the form of heat.

(167) In the implementation of FIG. 14A optical surfaces 367 and 368 were used in conjunction with a light reflector 369 to direct light emitted from the end emitting fiber 302 to a target location inside the proximal connector 304. In addition, each of the optical surfaces 367 and 368 were internal to the proximal connector 304. In the implementations of FIGS. 20A and 21A optical surfaces are used to direct light emitted by the end emitting fiber 301 to its target location without the use of a light reflector. Moreover, in the implementation of FIG. 21A at least some of the optical surfaces reside on and/or in an insert residing in a side cavity of the proximal connector 304. According to some implementations one or more or all of the optical surfaces has an RMS surface roughness of less than 180 angstroms.

(168) In each of the implementations disclosed and contemplated herein, when an end emitting fiber is used to deliver bacterial disinfecting light to a target location the end emitting fiber may be equipped with an end cap.

(169) FIG. 19D illustrates an implementation in which an end cap 651 is attached to the end of an end emitting fiber 302 and used to deliver light 659 directly to the light reflector 369 of end cap 306 without the use of an optical surface. That is, no structure exists between the end of the end cap and the light reflector 369 of the end cap 306. This arrangement results in a more simple design and can increase the efficiency by which light is delivered to its target location inside the proximal connector 304 and/or tubular body 303 of the infusion shaft.

(170) Each of the implementation of FIGS. 20A and 21A is similar to that of FIG. 14A except that no light reflector 369 is provided on the cap 306. Instead, to reflect the light these implementations use one or more optical surfaces on which total internal reflection occurs. These types of optical surfaces are also referred to herein as “total internal reflection optical surfaces”.

(171) Total internal reflection is the phenomenon which occurs when a propagated wave strikes a medium boundary at an angle larger than a particular critical angle normal to the incident surface. If the refractive index is lower on the opposing side of the boundary and the incident angle is greater than the critical angle, the wave cannot pass through and is entirely internally reflected. The critical angle is the angle of incidence above which the total internal reflection occurs. This is particularly common as an optical phenomenon, where light waves are involved.

(172) When a wave reaches a boundary between different materials with different refractive indices, the wave will in general be partially refracted at the boundary surface, and partially reflected. However, if the angle of incidence is greater (i.e. the direction of propagation is closer to

being parallel to the boundary) than the critical angle—the angle of incidence at which light is refracted such that it travels along the boundary—then the wave will not cross the boundary, but will instead be totally reflected back internally. This can only occur when the wave in a medium with a higher refractive index reaches a boundary with a medium of lower refractive index. For example, it will occur with light reaching air from plastic, but not when reaching plastic from air. (173) In the context of the present application, the term “reflector” and “light reflector” do not encompass a total internal reflection optical surface, but instead include polished surfaces, mirrors, metals and the like that reflect light regardless of the incident angle.

(174) Like the implementation of FIG. 14A, in the implementation of FIG. 20A the proximal connector 304 may be made of a plastic material. Generally, plastics have an index of refraction of between about 1.4 to about 1.7 as compared to air that has an index of refraction of 1.0 and saline that has an index of refraction of 1.33. In the implementation of FIG. 20A there are first, second, third and fourth optical surfaces 374-377 used to direct light from the end emitting fiber 302 to a target location inside the infusion shaft 300.

(175) The second and third optical surfaces 375 and 376 are each bounded on one side by the plastic material that forms the proximal connector 304 and on the other side by air in the proximal connector's manufactured state. In use, however, the internal cavity of the proximal connector may be occupied by a sterile saline solution. Thus, according to some implementations sterilization with the bacterial disinfecting light occurs when the proximal connector is filled with a saline solution. In such instances, the third optical surface 376 is bounded on one side by the plastic material (incidence medium) that forms the proximal connector 304 and on the other side by a saline solution. In regard to the second optical surface 375, a cavity 378 that is partially defined by the second optical surface is provided in a side of the proximal connector and filled with air. In addition, the angle of inclination of each of the optical surfaces 375 and 376 is chosen so that the incident angle at which light impinges on the surfaces is greater than the critical angle, the critical angle being the inverse sine of the ratio of the index of refraction of air or saline over the index of refraction of the plastic that forms the proximal connector. With the index of refraction of plastics ranging between about 1.4 to about 1.7, according to implementations where the optical surface is bounded on one side by air, the critical angle is greater than about 36 degrees to greater than about 46 degrees depending on the specific index of refraction of the plastic that is used in the construction of the proximal connector. According to implementations where the optical surface is bounded on one side by saline, the critical angle is greater than about 51 degrees to greater than about 72 degrees depending on the specific index of refraction of the plastic that is used in the construction of the proximal connector. Thus, according to some implementations optical surface 375 is configured to be impinged upon by the end emitted bacterial disinfecting light at an incident angle that is greater than about 36 degrees to about 46 degrees, and optical surface 376 is configured to be impinged upon by the end emitted bacterial disinfecting light at an incident angle that is greater than about 51 degrees to about 72 degrees.

(176) With reference to FIG. 20B, the path traveled by the light emitted by the end emitting fiber is shown. Like the implementation of FIG. 14A, a gap may exist between the end of fiber 302 and the end wall of the end emitting fiber lumen 310. In any event, the end wall 362a of lumen 362 constitutes the first optical surface 374 through which the light emitted by the fiber 302 passes. The gap may be filled with an index matching gel or adhesive that has an index of refraction between that of the core of fiber 302 and that of the plastic that forms the proximal connector 304. The light path 379 between the end of the fiber 302 and its target location involves a passing of the light through the first and fourth optical surface 374 and 377 and a total internal reflection on each of optical surfaces 375 and 376 as shown in FIG. 20B. In the context of the implementation of FIGS. 20A and 20B, each of optical surfaces 374 and 377 is a refractive optical surface and each of optical surfaces 375 and 376 is a total internal reflection optical surface.

(177) As explained above, a cavity 378 filled with air is formed in a side of the proximal connector

304. According to some implementations the cavity is open to the environment as shown in FIGS. **20A** and **20B**, whereas in other implementations the cavity **378** is closed to prevent any contaminants from entering the cavity after the cavity has been filled with purified air. According to some implementations closure of the cavity **378** is achieved by wrapping the outer surface of the proximal connector **304** with a heat shrink and then heat shrinking it tightly about the outer surface. According to some implementations, the heat shrink wrapping has light reflective properties that act to contain disinfecting light inside the proximal connector in a manner like that described above.

(178) The formation of the optical surfaces internal to the proximal connector **304** can be difficult. This is because pins with ends having optical quality finishes are used in the formation of the optical surfaces during a molding process. During the molding process the end of the pins are strategically located so that the molten plastic forms over the ends of the pins. Because the end of the pins have an optical quality finish, the plastic surfaces formed by the pins will also have an optical quality finish when the plastic solidifies.

(179) A problem associated with this method of forming the optical surfaces is that the desired location of the ends of the pins is not always easily accessible, and at times is not accessible at all. The implementation of FIGS. **21A** and **21B** addresses this issue with the use of an insert **380** that is manufactured separately from the proximal connector **304**. As shown in the figures, the insert **380** is affixed inside a cavity **381** located in a side of the proximal connector **304**. The system includes first, second, third and fourth optical surfaces **382-385**, of which the first, second and third optical surfaces **382-384** are located in or on the insert **380** and total internal reflection occurs at the second and third optical surfaces **383** and **384**. The end of the end emitting fiber **302** is located inside a lumen **388** of the insert **380**, and according to some implementations the insert **380** is constructed of the same material that forms the proximal connector **304**. The end wall of the insert lumen **388** constitutes the first optical surface **382** through which light passes toward the second optical surface **383**. Optical surfaces **383** and **384** are each bounded on one side by the plastic material that forms the insert **380** and on the other side by air **386**. In addition, the angle of inclination of each of the optical surfaces **383** and **384** is selected so that the incident angle at which the light impinges on these surfaces is greater than the critical angle, which in this case is the inverse sine of the ratio of the index of refraction of air over the index of refraction of the plastic that forms the insert **380**.

(180) With reference to FIG. **21B**, the path **388** traveled by the light emitted by the end emitting fiber **302** is shown. Like the implementations of FIGS. **14A** and **20A**, a gap may or may not exist between the end of fiber **302** and the end wall of the insert lumen **388**. When a gap is provided it may be filled with an index matching gel or adhesive that has an index of refraction between that of the fiber core and that of the material that forms the proximal connector **304**. The light path **389** between the end of the fiber **302** and its target location involves a passing of the light through the first and fourth optical surface **382** and **385** and a total internal reflection on each of optical surfaces **383** and **384** as shown in FIG. **21B**. As shown in FIGS. **21A** and **21B**, the insert **380** resides inside a cavity **381** formed in the side of the proximal connector **304**. The shapes of the insert **380** and cavity **381** are selected so that air gaps **386** exists at the boundary of each of the second optical surface **383** and third optical surface **384**. According to some implementations the cavity **381** is open to the environment as shown in FIGS. **21A** and **21B**, whereas in other implementations the cavity **381** is closed to prevent any contaminants from entering the cavity after the cavity has been filled with purified air.

(181) In the context of the implementation of FIGS. **20A** and **20B**, each of optical surfaces **382** and **385** is a refractive optical surface and each of optical surfaces **383** and **384** is a total internal reflection optical surface.

(182) In the exemplary implementation of FIG. **3** there exists eight optical fibers. These include the radially emitting fiber **205** and imaging fiber **207** of the main shaft **200** and also the radially

emitting fibers **301** and end emitting fibers **302** associated with each of the three infusion shafts **300**. As noted above, there may be fewer or more than three infusion shafts **300**. In addition, according to some implementations one or more of the infusion shafts **300** may possess only one of a radially emitting fiber **301** and an end emitting fiber **302** as exemplified above in the description of FIGS. **19A** and **19B**. Further, according to some implementations the main shaft **200** may include only one of a disinfecting radially emitting fiber **205** and an imaging fiber **207**. Moreover, one or more of the main shaft **200** and infusion shafts **300** may not be provided with any type of optical fiber. As such, it is evident that any of a variety of implementations contemplated herein may include any combination of fewer or more than eight optical fibers. With this in mind, the forthcoming description is directed to the implementation of FIG. **3**.

(183) Light may be delivered to the various optical fibers residing in the main shaft **200** and infusion shafts **300** in a number of ways. For example, the disinfecting radially emitting fibers **205** and **301** may each be optically coupled to a common first laser, the disinfecting end emitting fibers **302** may each be optically coupled to a common second laser, and the imaging fiber **207** may be optically coupled to a third laser as shown in FIG. **22C**. In this manner, the type of light delivered to each of the three types of fibers may be tailored to its intended function. The types of light may be distinguished, for example, by their wavelength and optical power. However, according to some implementations all of the optical fibers may be optically coupled to a single common laser as shown in FIG. **22A**, while in other implementations each optical fiber is optically coupled to its own individual laser as shown in FIG. **22B**. In this latter case the type of light delivered to each of the optical fibers may be individually tailored to its intended function like that discussed above in the implementation of FIG. **22C**.

(184) In the description that follows light is routed to the optical fibers residing in the main shaft **200** and infusion shafts **300** via the CVC hub **400**. However, in other implementations one or more or all of the optical fibers enter their respective lumens inside the main shaft and infusion shaft without having passed through the hub **400**. An advantage of running the optical fibers through the hub **400** as shown in FIG. **3** is that, other than the umbilical cord **500**, there are no other optical fibers residing outside the CVC **100** to interfere with a health care clinician's access to the various components of the CVC. As such, existing CVC clinical practices may be followed with little or no change.

(185) FIG. **23** is a top view of an optical fiber umbilical cord **500** that is connectable to the hub **400** of the CVC **100**. FIG. **24A** is a perspective view of an end of the umbilical cord **500**. FIG. **24B** is an end view of the distal end of the umbilical cord. The umbilical cord **500** is configured to transport light that originates from a one or more lasers to each of the optical fibers residing in the main shaft **200** and infusion shafts **300** of the CVC **100**. The umbilical cord **500** originates at a proximal optical connector **501** that has eight ports that receive and direct light to eight transport fibers **503** that run the length of the cord **500** with all but three of the transport fibers terminating at a location just distal to a hub connector **502**. The hub connector **502** facilitates a physical connection of the distal end portion of the cord **500** to the hub **400** which will be discussed in more detail below. In the implementation of FIGS. **23** and **24A-B**, the transport fibers **503** located inside the cord **500** comprise outer jackets and are disposed within the lumen of an elastomeric sheath. According to some implementations the fibers are located in Kevlar strength members located inside the sheath **504**. The sheath **504** originates at the distal end of connector **501** and terminates at or just distal to the hub connector **502**.

(186) In the implementation of FIG. **3**, five of the optical fibers that reside in the main shaft **200** and infusion shafts **300** are radially emitting fibers. These are fibers **205** and **207** of the main shaft **200** and fiber **301** in each of the three infusion shafts **300**. The remaining three optical fibers are the end emitting fibers **302** residing in the three infusion shafts **300**.

(187) Distal to the hub connector **502** each of the transport fibers **503** is devoid of an outer jacket.

(188) Five of the transport fibers **503** terminate a short distance distal to the hub connector **502** and

are subsequently physically and optically coupled to respective radially emitting fibers **205** and **207** of the main shaft **200** and radially emitting fibers **301** of the three infusion shafts **300** by the use of fiber optic couplers **505**. In the implementation of FIGS. **23** and **24A-B** the end emitting fibers **302** of the infusion shafts **300** are contiguous with the transport fibers **503**, but without the outer jacket. As will be explained in detail below, fibers **205**, **207**, **301** and **302** are routed to their respective lumens **204**, **206**, **309** and **310** inside the main shaft **200** and infusion shafts **300** via channels located in the hub **400**.

(189) As shown in the figures, according to some implementations, all or a majority of the fibers **205**, **207**, **301** and **302** extending from the umbilical **500** into the hub **400** are devoid of a jacket and comprise only a core/cladding in the radial emitting case and a core/cladding/buffer in the end emitting case. This arrangement significantly reduces the profile of the optical fibers and also the profile of the components into which they are incorporated.

(190) Proximal and distal strain reliefs **506** and **507** are respectively provided at the junction of the connector **501** with sheath **504** and at the junction of the hub connector **502** with sheath **504** to guard against the transport fibers **503** from breaking inside the umbilical cord at these junctions. In conjunction with or in lieu of providing strain reliefs, the sheath **504** may be structured and/or made of a material that prevents the sheath from bending beyond a minimum bending radius of one or more of the transport fibers **503**. The material may be, for example, a high durometer elastomer and/or a reinforced elastomer embedded with one or more stiffening elements (e.g. a coil, mandrel, braiding, Kevlar strength members etc.).

(191) FIG. **22A** illustrates an implementation of a laser system **520** that is configured to deliver light from a single laser **521** to the proximal optical connector **501** of the fiber optic umbilical cord **500**. The system includes a controller **522** that is connected to a user interface **523**. The system includes a laser driver **524** under the control of the controller **522**. In response to instructions received from the controller **522**, the driver **524** controls the operation of the laser **521**. A user interface **523**, such as a keypad or touchscreen, may be provided to communicate to the controller **522** user input instructions. The user input instructions may cause the controller **522** and driver **524** to operate the laser **521** in a user desired manner. For example, the user input instructions may be used to alter the wavelength and/or optical power of the light emitted by the laser **521**. According to some implementations the power level delivered to an optical fiber is regulated to provide a dose of disinfecting light that both kills bacteria and generates an amount of heat that does not result in patient discomfort (e.g. maintaining the CVC components that contact the patient below 104° F.). According to some implementations the power level of the light delivered is maintained constant whilst the power is turned on and off to provide the desired dose of disinfecting light. According to some implementations the optical power level of the light emitted from laser **521** is sufficient to deliver a desired dose like those discussed above. Patient comfort is maintained by ensuring the maximum irradiance impinging upon the skin is less than 200 mW/cm^{sup.2}.

(192) Light emitted by the laser **521** is transported by an optical fiber **525** (e.g. multimode fiber) to a beam splitter **526** that splits the laser beam emitted by the laser **521** into eight separate laser beams. The eight laser beams are individually transported via optical fibers **527** (e.g. multimode fibers) to a breakout box **528** that is configured to direct each of the eight laser beams into eight separate ports of an optical connector **529**. The optical connector **529** is in turn connected to a proximal connector **531** of an eight channel patch cord **530** that possesses eight transport fibers (e.g. multimode fibers). The transport fibers of the patch cord **530** deliver the eight laser beams to the proximal optical connector **501** of the fiber optic umbilical cord **500** via a distal connector **532**.

(193) According to some implementations the optical power of each of the laser beams transported by optical fibers **527** is sufficient to deliver a desired dose like those discussed above. In instances when light is to be delivered to both radially emitting fibers and end emitting fibers, each of the radially emitting fibers receive a larger proportion of the optical power than each of the end emitting fibers.

(194) According to some implementations two or more lasers (e.g. three lasers) may be combined to generate a common laser beam that passes through optical fiber **525**.

(195) According to each of the laser systems disclosed herein, when light is to be delivered to an imaging fiber, the laser system may possess a separate laser dedicated to delivering light only to the imaging fiber as in laser system **550** of FIG. **22C**.

(196) FIG. **22B** illustrates an implementation of a laser system **540** that is configured to deliver light from eight individual lasers **542a-h** to the proximal optical connector **501** of the fiber optic umbilical cord **500**. The system includes a controller **522** that is connected to a user interface **523**. The system includes an eight channel laser driver **541** that is under the control of the controller **522**. In response to instructions received from the controller **522** the eight channel driver **541** individually controls the operation of the lasers **542a-h**. A user interface **523**, such as a keypad or touchscreen, may be provided to communicate to the controller **522** user input instructions. The user input instructions may cause the controller **522** and eight channel driver **541** to operate the laser **521** in a user desired manner. For example, the user input instructions may be used to separately alter the wavelength and/or optical power of the light emitted by the lasers **542a-h**, or to turn the lasers on and off. The ability to separately control the lasers allows them to be activated at different times and/or with different power levels in a way that optimizes the efficacy of the disinfecting process and/or control the amount of heat generated in the CVC to prevent patient discomfort. To this end, different parts of the CVC may be disinfected at different times and/or at different power levels. According to some implementations the delivery of disinfecting light to an optical fiber is triggered by an event. For example, one or more components of the CVC may be equipped with a motion or temperature sensor that detects when a part of the CVC has been moved or touched. According to such implementations the sensor may communicate with the controller **522** to cause one or more of the lasers to be activated upon the occurrence of an event.

(197) Light emitted by the lasers **542a-h** is transported by optical fibers **544** (e.g. multimode fibers) to a breakout box **545** that is configured to direct each of the eight laser beams into eight separate ports of an optical connector **546**. The optical connector **546** is in turn connected to a proximal connector **531** of an eight channel patch cord **530** that possesses eight transport fibers (e.g. multimode fibers). The transport fibers of the patch cord **530** deliver the eight laser beams to the proximal optical connector **501** of the fiber optic umbilical cord **500** via a distal connector **532**.

(198) FIG. **22C** illustrates an implementation of a laser system **550** that is configured to deliver light from three lasers **552a-c** to the proximal optical connector **501** of the fiber optic umbilical cord **500**. The system includes a controller **522** that is connected to a user interface **523**. The system includes a three channel laser driver **551** under the control of the controller **522**. In response to instructions received from the controller **522** the driver **551** controls the operation of the three lasers **552a-c**. A user interface **523**, such as a keypad or touchscreen, may be provided to communicate to the controller **522** user input instructions. The user input instructions may cause the controller **522** and driver **551** to individually operate the lasers **552a-c** in a user desired manner. For example, the user input instructions may be used to individually alter the wavelength and/or optical power of the light emitted by each of the lasers **552a-c**, or to turn the lasers on and off. As stated above, the ability to separately control the lasers allows them to be activated at different times and/or with different power levels in a way that optimizes the efficacy of the disinfecting process and/or control the amount of heat generated in the CVC to prevent patient discomfort.

(199) Light emitted by the laser **552a** is transported by an optical fiber **556a** (e.g. multimode fiber) to a beam splitter **553a** that splits the laser beam emitted by the laser **552a** into four separate laser beams to be associated with the disinfecting radially emitting fiber **205** of the main shaft **200** and the disinfecting radially emitting fibers **301** of the three infusion shafts **300**. The four laser beams are individually transported via optical fibers **557a** (e.g. multimode fibers) to a breakout box **554** that is configured to direct each of the four laser beams into four separate ports of an optical connector **555**. Light emitted by the laser **552b** is transported by an optical fiber **556b** (e.g.

multimode fiber) to a beam splitter **553b** that splits the laser beam emitted by the laser **552b** into three separate laser beams to be associated with the disinfecting end emitting fibers **302** of the three infusion shafts **300**. The three laser beams are individually transported via optical fibers **557b** (e.g. multimode fibers) to a breakout box **554** that is configured to direct each of the three laser beams into three separate ports of the optical connector **555**. Light emitted by laser **552c**, that is to be associated with the imaging fiber **207** of the main shaft **200**, is transported by an optical fiber **556c** (e.g. a multimode fiber) directly to the breakout box **554** which is configured to direct the laser beam into a port of the optical connector **555**. The optical connector **555** is in turn connected to a proximal connector **531** of an eight channel patch cord **530** that possesses eight transport fibers (e.g. multimode fibers). The transport fibers of the patch cord **530** deliver the eight laser beams to the proximal optical connector **501** of the fiber optic umbilical cord **500** via a distal connector **532**. (200) According to some implementations each of lasers **522a** and **522b** comprise blue lasers and laser **552c** is a red laser.

(201) FIG. **25** shows a top perspective view of the various parts that from the hub **400** according to one implementation. The parts include a bottom tray **401**, a mid-tray **402**, a cover **403** and a top gasket **404**, of which enlarged perspective views are respectively provided in FIGS. **26-29**. The bottom tray **401** and mid-tray **402** include features for supporting the proximal end of the main shaft **200** and the distal ends of the infusion shafts **300** relative to one so that the working lumens **308** of the infusion shafts and the working lumens **201a-c** of the main shaft may be connected with the use of conduits located inside the hub **400**. The bottom tray **401** and mid-tray **402** also include features that define one or more channels for strategically routing the optical fibers through the hub **400** in a manner that protects the optical fibers against breakage. The channels that possess the radially emitting fibers may also run above, below or to the side of the conduits inside the hub that join the working lumens **308** of each of the infusion shafts **300** to the working lumens **201a**, **201b** and **201c** of the main shaft **200**. The fiber channels may be formed by any of a variety of structures including, but not limited to, continuous wall segments, spaced-apart posts, etc. In channels that have one or more bends along their length, to protect against breakage of the optical fibers, the bends are constructed to prevent a bending of the optical fiber housed therein beyond its minimum bending radius. For example, each of the one or more bends may have a radius of curvature that is equal to or greater than the minimum bending radius of the optical fiber.

(202) As shown in FIG. **26**, according to some implementations the bottom tray **401** has a plurality of channels formed therein. Each of channels **405a-c** is configured to hold one of the infusion shafts **300**. The depth and width dimensions of the channels **405a-c** may respectively correspond to the length and width dimensions of the infusion shaft key portions **311**. A receptacle **406** in the bottom tray **401** is configured to accept the hub connector **502** of the optical fiber umbilical **500** in a manner that locks the proximal connector onto the bottom tray. Distal to the receptacle **406** is an elongate channel **407** that guides the optical fibers that emanate from the umbilical **500** into the hub **400**. Distal to channel **407** there resides a raised platform **408** that includes a top surface **408a** and a curved wall portion **408b**. (It is important to note that the top surface **408a** need not be a planar/level surface) A channel **409**, which is at least partially formed by the curved wall portion **408b** of the raised platform **408**, communicates fiber channel **407** to fiber channels **410a-c**. Each of fiber channels **410a-c** is respectively located distal to channels **405a-c** and assists in guiding the radially emitting fiber **301** and/or end emitting fiber **302** into their respective lumens **309** and **310** inside the key portions **311** of the infusion shafts **300**.

(203) FIG. **30** illustrates a perspective view of a partially assembled hub according to one implementation with the mid-tray **402** being attached to the bottom tray **401**. A distal end **390a-c** of each of the infusion shafts **300** extends distally from the mid-tray **402**. In the implementation of FIG. **30** there exist eight optical fibers consistent with the embodiment of FIG. **3** described above. According to the implementation of FIG. **30** all of the fibers **205**, **207**, **301** and **302** extend through fiber channels **407** and **409** before they branch apart. Six of the fibers (that is fibers **301** and **302** for

each of the three infusion shafts **300**) pass to the infusion shafts **300** via fiber channels **410a-c** while two of the fibers (that is fibers **205** and **207** of the main shaft **200**) pass to the main shaft **200** via a fiber channels **412** and **413**. As shown in FIG. **30**, the various fiber channels may be disposed at different elevations within the hub **400**. This provides greater flexibility in laying out the optical fiber pathways.

(204) FIGS. **27A** and **27B** respectively show top and bottom perspective views of the mid-tray **402** according to one implementation. The mid-tray includes grooves **413a-c** and **414** that each extend between its proximal and distal ends **418** and **419**, respectively. As shown in FIG. **30**, when the infusion shafts **300** are assembled on the bottom tray **401** and in the mid-tray **402** the distal end portion **390a-c** of each of the infusion shafts protrudes distal to the distal side **419** of the mid-tray. According to some implementations the portions of the tubular body **303** of each of the infusion shafts that form the working lumens **308** (e.g. semi-circular region of the infusion shaft) fit tightly within their respective grooves **413a-c**. In some instances the fit between the infusion shafts **300** and the grooves **413a-c** produces a liquid-tight seal at their interface along at least a portion of the length of the grooves **413a-c**. According to some implementations the mid-tray **402** also includes a receptacle **417** that houses at least a portion of the hub connector **502** of the optical fiber umbilical **500**.

(205) During an assembly of the hub **400** a portion of the main shaft **200** may be supported on a cradle **411** positioned atop the raised platform **408**. According to some implementations the cradle **411** has a grooved external surface **411a** that conforms to the external surface of the main shaft. When positioned on the cradle **411**, as shown in FIG. **30**, the elongate proximal end portion **220** of the main shaft **200** extends into the curved channel **413** of the mid-tray **402** where fibers **205** and **207** enter the proximal end **221** of the elongate proximal end portion **220**.

(206) Assembly of the CVC **100** may occur in any of a variety of steps to which the following assembly method represents only but one example. At step one the hub connector **502** of the fiber optic umbilical cord **500** is fixed inside the receptacle **406** of the bottom tray **401**. The radially emitting fibers **301** and end emitting fibers **302** associated with the infusion shafts **300** are then placed inside the lumens **309** and **310** of their respective infusion shafts. This step may include fixing the distal end of the end emitting fibers **302** inside the proximal connectors **304** of the infusion shafts **300** with, for example, an index matching adhesive as discussed above. With the fibers **301** and **302** properly positioned inside their respective lumens inside the infusion shafts **300**, the distal end portion of each of the infusion shafts is positioned on the bottom tray **401**. According to some implementations this includes aligning each of the distal end portions of infusion shafts with channels **405a-c** in the bottom tray **401** with each of their key portions **311** respectively residing in one of said channels. The key portions **311** may be pressed-fit into the channels **405a-c** and/or secured inside said channels with the use of an adhesive.

(207) With the proximal end portions of the infusion shafts **300** in place on the bottom tray **401** the six fibers **301** and **302** are routed through the bottom tray fiber channels **407**, **409** and **410a-c**.

(208) With the infusion shafts **300** and fibers **301** and **302** in place, the mid-tray **402** is positioned over and attached to a proximal end portion of the bottom tray **401**. When the mid-tray **402** is attached to the bottom tray **401** the distal end portions of the infusion shafts reside inside their respective grooves **413a-c** inside the mid-tray **402** as depicted in FIG. **30**. After the mid-tray **402** is positioned on the bottom tray **401**, the radially emitting fiber **205** and imaging fiber **207** are routed through fiber channels **407**, **409**, **412** and **413** and into their respective lumens **204** and **206** in the main shaft **200**.

(209) As discussed above, according to some implementations one or more or all of the curves of the structures that form the fiber channels (e.g. channels **407**, **409**, **410a-c** and **413**) are endowed with a radius of curvature that is no less than the minimum bending radius of one or more of the fibers that pass through them.

(210) Because a later step involves a casting of a urethane or other polymeric material to

encapsulate the hub, with the infusion shafts and optical fibers in place the cover **403** is situated to reside above the fiber channels **407**, **409**, **410a-c**, **412** and **415** formed in the bottom tray **401** and mid-tray **402** as shown in FIG. **31**. The cover **403** serves to isolate the optical fibers from the casting material so that the optical fibers maintain a freedom of movement inside the hub when the casting is complete. The cover **403** includes an opening **441** through which the cradle **411** of the bottom tray protrudes. The cover **403** also includes opening **442a-c** through which the distal ends **390a-c** of the infusion shafts **300** respectively protrude.

(211) With the cover **403** in place to protect the optical fibers as shown in FIG. **31**, the main shaft **200** is positioned on the cradle **411** atop the raised platform **408** with at least a portion of the length of its elongate proximal end portion **220** placed inside the fiber channel **413**. According to some implementations the main shaft **200** is positioned on the cradle **411** so that the proximal opening of the working lumen **201a** resides proximal to the raised platform **408** as shown in FIG. Positioning the proximal end of the working lumen **201a** of the main shaft proximal to the raised platform **408** advantageously shortens the path taken to couple the working lumens **308** of the infusion shafts to the infusion shafts **201a-c** of the main shaft **200**. This positioning also facilitates a routing of at least a portion of the length of the conduits that interconnect the infusion shafts to the main shaft above the fiber channels **410a-c** so that these lengths of the conduits may be irradiated with bacterial disinfecting light emitted by the radially emitting fibers **301** that reside in fiber channels **410a-c**.

(212) As shown in FIG. **30**, one or more or all of the fibers **205**, **207**, **301** and **302** is routed in the hub **400** so as not to be taut along at least a portion of the structures that form the fiber channels located in the bottom tray **401**. This provision of slack in the one or more optical fibers inside the hub guards against excessive tensile forces being applied to the fibers when the main shaft **200** and infusion shafts **300** are bent or pulled in tension as a result of the slack being taken up inside the hub when the shafts are bent. This is particularly important where at least a portion of the optical fiber is longitudinally fixed inside the main shaft or infusion shaft, like with certain implementations of the imaging shaft **205** and end emitting fibers **302** discussed above.

(213) With the cover **403** in place, the top gasket **404** is placed atop the periphery of the cover **403** to hold the cover in place prior to the casting step. The gasket **404** includes at its distal end a groove **445** through which the main shaft **200** passes. As shown in FIGS. **25-31**, holes and posts located about the periphery of the bottom tray **401**, mid-tray **402**, cover **403** and top gasket **404** facilitate an aligning and coupling of the components prior to the casting step.

(214) As discussed above, conduits are used to fluidly connect the working lumens **308** of the infusion shafts to the respective working lumens **201a-c** of the main shaft **200**. According to some implementations the conduits are formed during the casting process mentioned above. According to such implementations when the hub is in a partially assembled state as shown in FIG. **31**, an elongate flexible beading/mandrel **450** (see FIG. **32**) is inserted through each of the working lumens **308** of the infusion shafts **300** and into the respective working lumens **201a-c** of the main shaft **200**. With the beadings in place the entirety of the assembly is cast with a polymeric material, such as a urethane. The casting is performed so that the casting material totally encapsulates those portions of the beadings **250** that extend between the distal end of the infusion shafts and proximal end of the main shaft. Upon the casting material having cured, the beadings **450** are removed so that flow channels encapsulated inside the casting exist between the working lumens of the infusion shafts **300** and main shaft **200**. FIG. **3** shows the external surface of the hub **400** when the casting step is complete.

(215) According to other implementations, as shown in FIG. **33**, the conduits **460a-c** that connect the working lumens of the infusion shafts and main shaft are not cast, but are instead physically and fluidly coupled to their respective working lumens before the casting step. According to such implementations the conduits **460a-c** may be fluidly sealed to the working lumens of the infusion shafts and main shaft prior to the casting process. According to other implementations the

interfaces between the conduits **460a-c** and the working lumens of the infusion shafts and main shaft are sealed with the casting material during the casting process.

(216) According to some implementations the casting material and the cover **403** of the hub **400** each comprises a material that is transparent or translucent to the light emitted by the disinfecting radially emitting fibers **205** and **301**. According to other implementations the casting material, mid-tray **402** and cover **403** of the hub **400** each comprises a material that is transparent or translucent to the light emitted by the disinfecting radially emitting fibers **205** and **301**. According to yet other implementations the casting material, bottom tray **401**, mid-tray **402** and cover **403** of the hub **400** each comprises a material that is transparent or translucent to the light emitted by the disinfecting radially emitting fibers **205** and **301**. By virtue of their transparency or translucence, these components of the hub enable light emitted by the light disinfecting radially emitting fibers to disinfect not only the infusion shaft **300** and main shaft **200** lumens, but to also internally flood the hub with bacterial disinfecting light to also effectuate a disinfecting of the hub itself, including its outer surfaces.

(217) As shown in FIG. **34**, according to some implementations light reflectors **453** and **454** may be affixed to the top and bottom surfaces **451** and **452** of the hub **400** in order to minimize the loss of disinfecting light out of the hub. A trapping of the bacterial disinfecting light inside the hub enhances the light exposure on the internal components of the hub. This increases the bacteria killing effectiveness of the disinfecting light emitted into the hub. It also reduces the amount of time and/or the power needed to effectuate a disinfecting of the hub. According to other implementations, the casting material used to cast the hub may be impregnated with light reflecting elements that act to direct light into the hub. According to other implementations the casted hub **400** is wrapped with a material that has an internal light reflective surface. According to one implementation the material comprises shrink wrap that is shrunk about the external surfaces of the hub. According to yet other implementations, one or more of the fiber channels located in the bottom tray **401** have surfaces that are coated with a light reflective material, such as, for example, a light reflective paint.

(218) According to some implementations disinfecting light is delivered into the CVC only by way of use of end emitting fibers. Thus, unlike the implementation of FIG. **30**, each of the infusion shafts **300** and main shaft **200** of the implementation of FIGS. **35A-38** is devoid of a radially emitting fiber and a radially emitting fiber lumen.

(219) In the implementation of FIGS. **35A-38** the disinfecting light is delivered into each of the infusion shafts **300** at two locations. The first location is in the proximal connector **304** of the infusion shaft **300** in which disinfecting light is delivered in a manner consistent with what has been described above with one or more optical surfaces and/or one or more light reflectors of the proximal connector being used to disperse disinfecting light emitted from an end emitting fiber **302** into at least a portion of the infusion shaft **300**. The second location is inside the working lumen **308** of the infusion shaft **300** near its distal end. This second location is located inside the hub **400**.

(220) Although not shown in the figures, the CVC of FIGS. **35A-38** may also include an imaging fiber **207** that runs along a length of the main shaft **200** as previously described.

(221) In the implementation depicted in FIGS. **35A** and **36** there are six optical fibers that extend from the fiber optic umbilical **500** into the hub **400**. The six optical fibers are comprised of three end emitting fibers **302** that deliver light to the proximal connectors **304** of the infusion shafts and three end emitting fibers **337a-c** that deliver light into the working lumens **308** of the infusions shafts inside the hub. The main shaft **200** and infusion shafts **300** are supported in the hub **400** in a manner consistent with that described above in conjunction with the implementation of FIG. **30**.

(222) FIG. **35B** shows the hub of FIG. **35A** without the optical fibers **302** and **337a-c** to better identify the fiber channels **407**, **409**, **410a-c**, **412**, **416**, **420**, **422**, **423a-b** and **427** running through the hub **400**. With reference to each of FIGS. **35A** and **36**, the end emitting fibers **302** that extend into the proximal connectors **304** of the infusion shafts **300** pass through the bottom tray **401** of the

hub via fiber channels **407**, **409** and a respective one of channels **410a-c**. End emitting fiber **337a** passes from the fiber optic umbilical **500** and into the fiber channel **423a** in the mid-tray **402** via fiber channels **407**, **409**, **427**, and **422**. Each of end emitting fiber **337b** and **337c** passes from the fiber optic umbilical **500** and into its respective fiber channel **423b** and **423c** in the mid-tray **402** via fiber channels **407**, **409** and **420**. In the implementation of FIGS. **35A** and **36** at least a portion of each of the end emitting fibers **337a**, **337b**, **337c** overlaps with another one of the end emitting fibers **337a**, **337b**, **337c** in one or both of the bottom tray **401** and mid-tray **402**.

(223) FIG. **38** shows a cross-section view of the hub **400** with a first of the infusion shafts **300** secured between the bottom tray **401** and mid-tray **402**. The end of the end emitting fiber **337a** rests inside channel **423a** facing an optical surface **426a** formed on an inner surface of the mid-tray **402**. In the implementation of FIG. **38**, the end emitting fiber **337a** is optically coupled with the optical surface **426a** by use of an index matching gel or adhesive **428**. According to one implementation an index matching adhesive is used to both secure the fiber **337a** inside the downward sloping channel **423a** and to optically couple the end emitting fiber to the optical surface **426a**. A recess **424a** inside the mid-tray **402** is positioned to reside over an opening **331** cut into the tubular body **303** of the infusion shaft **300**. The arrangement of the end emitting fiber **337a**, optical surface **426a**, recess **424a** and opening **331** results in bacterial disinfecting light **339** being transmitted from the end emitting fiber into the working lumen **308** of the infusion shaft **300** when disinfecting light is transmitted through the end emitting fiber. At least that portion of the mid-tray that resides between the distal end of optical fiber **337a** and recess **424a** is made of a material that is transparent or translucent to the disinfecting light **339**. According to some implementations the entirety of the mid-tray **402** is made of a material that is transparent or translucent to the disinfecting light **339**.

(224) End emitting fibers **337b** and **337c** may be arranged similar to fiber **337a** to effectuate a disinfecting of at least a portion of the working lumens **306** of the other infusion shafts **300**.

(225) Like with the implementation of FIG. **30**, one or more or all of the fibers **302** and **337a-c** is routed in the hub **400** so as not to be taut along at least a portion of the structures that form the fiber channels located in the bottom tray **401**. This provision of slack in the one or more optical fibers guards against excessive tensile forces being applied to the fibers when the infusion shafts **300** are bent. This is particularly important where at least a portion of the optical fiber is longitudinally fixed inside infusion shaft, like with certain implementations of the end emitting fibers **302** discussed above.

(226) Fluidly connecting the working lumens **308** of the infusion shafts **300** to the working lumens **201a-c** of the main shaft **200** may be accomplished in a manner similar to that described above in conjunction with the implementation of FIG. **30**. Like with the implementation of FIG. when the casting process is complete, the entirety of the hub **400** may be encapsulated by the casting material and assume an appearance like that shown in FIG. **3**.

(227) To protect against breaking the end emitting fibers, the bends of the fiber channels that contain them are constructed to prevent a bending of the optical fibers beyond their minimum bending radius. For example, one or more or all of the bends may have a radius of curvature that is equal to or greater than the minimum bending radius of the optical fiber. Also, as shown in FIG. **36**, the path of the optical fibers about the raised platform **408** may be selected to ensure that the optical fibers do not bend beyond their minimum bending radius. For this purpose some of the optical fibers may travel in a clockwise direction about the raised platform **408**, while others travel in a counter-clockwise direction about the raised platform. In the Implementation of FIGS. **35A-38** all the end emitting optical fibers travel in a clockwise direction about the raised platform **408** (as viewed from above the bottom tray **401**) except fiber **337a** that travels in a counter-clockwise direction about the raised platform. The provision of alternate pathways for the optical fibers to travel around the raised platform **408** increases design freedom, particularly in the layout of the bottom tray **410** and mid-tray **402** fiber channels. The provision of alternate pathways can also facilitate a smaller hub design to provide it with a footprint that is more comparable to the foot

print of a traditional CVC hub. As discussed above, by maintaining the disinfecting CVC design similar to traditional CVC designs established clinical practices may be followed.

(228) Light reflectors may be incorporated into, attached to or coated on the hub **400** in a manner consistent with that described above in conjunction with the implementation of FIG. **30**.

(229) FIG. **39A** illustrates a partially constructed hub **400** of a CVC according to another implementation. Like with the previously described implementations, the hub is configured to fluidly connect the working lumens **308** of the infusion shafts **300** to the working lumens **201a-c** of the main shaft **200**. The conduits that connect the working lumens may be like those discussed above in conjunction with the implementation of FIG. **30**.

(230) According to some implementations disinfecting light is delivered into the CVC only by way of use of end emitting fibers. As a result, according to some implementations each of the infusion shafts **300** and main shaft **200** is devoid of a radially emitting fiber and a radially emitting fiber lumen.

(231) In the implementation of FIG. **39A** the disinfecting light is delivered into one or more of the proximal connectors **304** of the infusion shafts **300** in a manner consistent with what has been described above with one or more optical surfaces and/or one or more light reflectors of the proximal connector being used to disperse disinfecting light emitted from an end emitting fiber **302** into at least a portion of the infusion shaft. Disinfecting light is also delivered out the ends **358** of other end emitting fibers **338a-c** to flood at least that region of the hub that contains the conduits connecting the working lumens of the infusion shafts **300** and main shaft **200**.

(232) According to some implementations the ends of end emitting fibers **338a-c** comprise end caps **358** that are optically coupled to the cores of the end emitting fibers. When the hub is fully constructed the ends of the end emitting fibers **336a-c** will reside in the casting material that envelops the hub. The use of the end caps **358** provides at least two advantages. The advantage is that by lowering the power density of the light that exits the fibers the light will be below the damage threshold of the casting material.

(233) As shown in FIG. **39D**, according to some implementations support fixtures **364a-c** are provided at or near the distal end of channels **421a-c** to assist in respectively fixing the ends of the end emitting fibers **338a-c** on the hub. As illustrated in FIGS. **39A** and **39C**, according to some implementations the core and cladding portion of the end emitting fibers **338a-c** are clamped inside the support fixtures **364a-c** with the end caps **358** being positioned to protrude from the face **402a** of the mid-tray **402**. According to other implementations the end caps **358** are instead held fixing inside the support fixtures **364a-c**.

(234) Although not shown, the CVC of FIG. **39A** may also include an imaging fiber **207** that runs along a length of the main shaft **200** as described above.

(235) In the implementation depicted in FIG. **39A** there are six optical fibers that extend from the fiber optic umbilical **500** into the hub **400**. The six optical fibers are comprised of three end emitting fibers **302** that deliver light to the proximal connectors **304** of the infusion shafts **300** and three end emitting fibers **338a-c** that deliver light into the region of the hub that contains the conduits connecting the aforesaid working lumens of the infusion shafts **300** and main shaft **200**. The main shaft **200** and infusion shafts **300** are supported in the hub **400** in a manner consistent with that described above in conjunction with the implementation of FIG. **30**.

(236) FIG. **39B** shows the hub of FIG. **39A** without the optical fibers **302** to better identify the fiber channels **407**, **409**, **410a-c**, **412**, **416**, **420**, **421a-c**, **422** and **427** running through the hub **400**. The end emitting fibers **302** that extend into the proximal connectors **304** of the infusion shafts **300** pass through the bottom tray **401** of the hub via fiber channels **407**, **409** and a respective one of channels **410a-c**. End emitting fiber **338a** passes from the fiber optic umbilical **500** and into the fiber channel **421a** in the mid-tray **402** via fiber channels **407**, **409**, **427**, and **422**. Each of end emitting fiber **338b** and **338c** passes from the fiber optic umbilical **500** and into its respective fiber channel **421b** and **421c** in the mid-tray **402** via fiber channels **407**, **409** and **420**. In the implementation of

FIG. 39A at least a portion of each of the end emitting fibers **338a**, **338b**, **338c** overlaps with another one of the end emitting fibers **338a**, **338b**, **338c** in one or both of the bottom tray **401** and mid-tray **402**.

(237) As shown in FIG. 39A, the distal end of each of the end caps **358** of the end emitting fibers **338a-c** faces distally toward the region where the conduits that join the working lumens of the infusion shafts and main shaft are to reside. Accordingly, disinfecting light emitted by the end emitting fibers **302** is directed on and through the conduits to effectuate a disinfecting of the same. For this purpose, the casting material that is ultimately used to encapsulate the hub and to form the connecting conduits inside the casting is transparent or translucent to the disinfecting light.

(238) Like with the implementation of FIG. 30, one or more or all of the fibers **302** and **338a-c** is routed in the hub **400** so as not to be taut along at least a portion of the structures that form the fiber channels located in the bottom tray **401**. This provision of slack in the one or more optical fibers guards against excessive tensile forces being applied to the fibers when the infusion shafts **300** are bent. This is particularly important where at least a portion of the optical fiber is longitudinally fixed inside the infusion shaft, like with certain implementations of the end emitting fibers **302** discussed above.

(239) Fluidly connecting the working lumens **308** of the infusion shafts **300** to the working lumens **201a-c** of the main shaft **200** may be accomplished in a manner similar to that described above in conjunction with the implementation of FIG. 30. Like with the implementation of FIG. when the casting process is complete, the entirety of the hub **400** may be encapsulated by the casting material and assume an appearance like that shown in FIG. 3.

(240) To protect against breaking the end emitting fibers, the bends of the fiber channels that contain them are constructed to prevent a bending of the optical fibers beyond their minimum bending radius. For example, one or more or all of the bends of the fiber channels may have a radius of curvature that is equal to or greater than the minimum bending radius of the optical fiber. The path of the optical fibers about the raised platform **408** in the bottom tray **401** may also be selected to ensure that the optical fibers do not bend beyond their minimum bending radius. For this purpose some of the optical fibers may travel in a clockwise direction about the raised platform **408**, while others travel in a counter-clockwise direction about the raised platform. In the implementation of FIG. 39A all the end emitting optical fibers travel in a clockwise direction about the raised platform **408** (as viewed from above the bottom tray **401**) except fiber **358a** that travels in a counter-clockwise direction about the raised platform. The provision of alternate pathways for the optical fibers to travel around the raised platform **408** increases design freedom, particularly in the layout of the bottom tray **401** and mid-tray **402** fiber channels. The provision of alternate pathways can also facilitate a smaller hub design to provide it with a footprint that is more comparable to the foot print of a traditional CVC hub. As discussed above, by maintaining the disinfecting CVC design similar to traditional CVC designs established clinical practices may be followed.

(241) Light reflectors may be incorporated into, attached to or coated on the hub **400** in a manner consistent with that described above in conjunction with the implementations of FIG. 30. For example, the casting material that surrounds the connecting conduits when the hub is encapsulated may have embedded therein one or more light reflecting elements that assist in directing the disinfecting light toward the conduits.

(242) As mentioned above, in certain instances radially emitted disinfecting light and end emitted disinfecting light may be delivered to certain parts of the CVC by use of a single optical fiber. In the implementations of both FIGS. 35A and 39A one or more of the end emitting fibers may be substituted with such a fiber to increase the surface area that is exposed to the disinfecting light. For instance, in the implementation of FIG. 39A each of fibers **302** may be substituted with a dual radial and end emitting fiber. This would result in a disinfection of the connecting conduits with the end emitted light as discussed above, and would also result in a disinfection of those portions of the

infusion shafts that reside below the mid-tray **402** with the radial emitted light. For this purpose, according to some implementations at least a portion of the length of the fiber channels **421a-c** located in the mid-tray **402** are aligned with the tubular body **303** of the infusion shafts **300** located below the mid-tray. Furthermore, because all of the optical fibers in one way or another pass across at least a portion of the main shaft **200** inside the hub **400**, the radial emitted light emanating from these fibers can also be used to cause a disinfecting of the main shaft inside the hub.

(243) In a modified implementation of the CVC of FIG. **35A** one or more of fibers **337a-c** is a dual radial and end emitting fiber and fibers **302** are end emitting fibers. In a like manner, in a modified implementation of the CVC of FIG. **39A** one or more of fibers **338a-c** is a dual radial and end emitting fiber and fibers **302** are end emitting fibers.

(244) FIG. **40** depicts a partially constructed hub **470** of a CVC according to another implementation wherein optical fibers **471** are routed through the hub by use of support fixtures **472** dispersed about and supported on the hub tray **473**. The hub is configured to support the distal ends of the infusion shafts **300** and the proximal end of the main shaft **200**. Upon the optical fibers **471** being properly routed into their respective lumens inside the infusion shafts **300** and/or main shafts **300**, the fibers are appropriately covered during a process in which the hub is encased by a casting material. With beadings/mandrels extending between the working lumens of the infusion shafts **300** and main shaft **200** as described above, the conduits that fluidly connect the working lumens are formed during the casting process. The support fixtures **472** are disbursed about the hub in a manner that prevents a bending of one or more or all of the optical fibers **471** beyond their minimum bending radius. In the implementations discussed above, one or more of the fiber channels may be substituted with one or more support fixtures **472** to guide the fibers to their desired destination and in a way that prevents them from bending beyond their minimum bending radius.

(245) According to one implementation the support fixture **472** includes an expandable elastomeric O-ring **475** that has a top opening/gap **476** through which the optical fibers **471** may be introduced into the O-ring opening **477**. The elastomeric O-ring is resiliently biased in the position shown in FIG. **41**.

(246) The O-ring is supported on a post **478** that is attached to the floor of the hub tray **473**. Notches **480** in the bottom portion of the O-ring produce hinges that allow the arms **475a** and **475b** of the O-ring to be flexed outward in opposite directions A and B to increase the size of the gap **476** when a downward force is applied to the protruding arms **479** located near the base of the O-ring. When the downward applied force to the arms **479** is released, the arms **475a** and **475b** of the O-ring attempt to return to their original position to lock the optical fibers within the opening **477**.

(247) As discussed above in conjunction with FIGS. **10B** and **10C**, according to some implementations one or more of the infusion shafts **300** are equipped with dual working lumens **308a** and **308b** that are separated by a longitudinal wall/septum **321**. In the aforesaid implementations the radially emitting fiber lumen **309** and end emitting fiber lumen **310** are located in the septum **321**.

(248) In the discussion that follows, apparatus and methods for forming conduits between the respective working lumens of the infusion shafts **300** to the working lumens of the main shaft are disclosed. A portion of the conduits are formed during a casting process like those described above, wherein a casting material is used to encapsulate the hub. During the casting process at least a portion of the length of the conduits is formed with the use of strategically placed mandrels that are enveloped in the casting material during the casting process and ultimately removed.

(249) According to some implementations the apparatus includes a hub **600** that is adapted to makes use of mandrels **604a**, **604b** to form conduits to connect the two working lumens **308a**, **308b** of a single infusion shaft **300** to a common working lumen inside the main shaft **200**. Although the discussion below is directed primarily to fluidly coupling the two working lumens of an infusion shaft **300c** to the working lumen **201c** of the main shaft **200**, it is appreciated that the working

lumens of the other infusion shafts **300a**, **300b** may be coupled with the working lumens **201a** and **201b** of the main shaft **200** in a similar manner.

(250) According to one implementation the hub **600** includes a tray **601** onto which are arranged at a proximal end thereof infusion shaft support fixtures **605a-c**. The proximal end of the main shaft **200** of the CVC resides inside a manifold **602** located at the distal end of the tray **601**. As best seen in FIG. **43**, the manifold includes internal conduits **606a**, **606b**, **607a**, **607b**, **608a** and **608b** that are fluidly coupled with the working lumens **201a-c** of the main shaft **200**. Internal conduits **606a** and **606b** are fluidly coupled with working lumen **201a**, internal conduits **607a** and **607b** are fluidly coupled with working lumen **201b**, and internal conduits **608a** and **608b** are fluidly coupled with working lumen **201c**.

(251) The proximal face of the manifold **602** includes mandrel support fixtures **610** and **612** that have multiple lumens (e.g. 2 lumens) fluidly coupled to the internal conduits of the manifold that are respectively coupled to the working lumens **201b** and **201c** of the main shaft **200**. Although not shown in the figures, the manifold **602** may also include a mandrel support fixture that has multiple lumens (e.g. 2 lumens) that communicate with the working lumen **201a** of the main shaft **200**.

(252) FIG. **44** shows a distal side of the manifold **602** that includes an opening **609** into which the proximal end of the main shaft **200** resides when the hub **600** is fully assembled. The opening **609** is in fluid communication with each of internal conduits **606a**, **606b**, **607a**, **607b**, **608a** and **608b**. According to some implementations the proximal end of the main shaft **200** is fixed inside the opening **609** by use of an adhesive. The proximal end of the main shaft may also be press-fit inside the opening **609**. In any event, each of the inlets of the working lumens **201a-c** are aligned with a respective pair of the internal conduits as shown in FIG. **43**.

(253) The mandrel support fixture **612** includes a proximal side **615**, a distal side **616** and two through lumens **614a** and **614b** that each extends between the proximal and distal sides. Through lumen **614a** is in fluid communication with manifold conduit **608a** and through lumen **614b** is in fluid communication with manifold conduit **608b**. According to some implementations the mandrel support fixtures **610** and **612** located on the sides of the manifold **602** each have sloped proximal sides **615**. For example, as shown in FIGS. **43** and **45B**, the outer side **613a** of the fixture **612** has a thickness **t1** and the inner side **613b** of the fixture **612** has a thickness **t2** that is less than the thickness of **t1**. The purpose of the sloped proximal side is discussed below. According to some implementations fixture **612** is attached to the manifold **602** by the use of posts **617** that protrude from its distal side **16** into openings (not shown) in the proximal side of the manifold **602**.

(254) As briefly mentioned above, a proximal end of the hub tray **601** includes infusion shaft support fixtures **605a-c**. Each of the infusion shaft support fixtures **605a-c** is configured to couple the distal end of an infusion shaft **300a-c** to the hub **600** and to also accommodate a passage of the mandrels from inside the infusion shafts into the hub **600**. When, for example, the mandrels **604a** and **604b** are in place during the manufacturing of the hub **600** as shown in FIGS. **42** and **46**, they extend through the working lumens **308a**, **308b** of the infusion shaft **300c**, through the infusion shaft support fixture **605c** and into the mandrel support fixture **612** located on the manifold **602**.

(255) FIGS. **47A** and **47B** respectively show the proximal and distal sides **622** and **624** of the infusion shaft support fixture **605c** according to one implementation. The fixture includes first and second through lumens **619a** and **619b** that extend between the proximal and distal sides **622** and **624**. Disposed between lumens **619a** and **619b** is a septum **626** through which two other through lumens **628a** and **628b** pass. According to some implementations the distal end of the infusion shaft **300c** is attached to the proximal side **622** of the fixture **605c** by use of a pair of wall segments **629a** and **629b** that protrude proximally from the septum **626**. The wall segments are spaced-apart so that a gap exists between them. Attachment of the proximal end of the infusion shaft **300c** is accomplished by inserting the septum **321** of the infusion shaft between the spaced-apart wall segments **629a** and **629b** so that the face of the shaft septum **321** abuts the proximal face of the fixture septum **626**. When the infusion shaft **300c** is attached to the fixture **605c** the radially

emitting fiber lumen **309** is axially aligned with lumen **628a** and the end emitting fiber lumen **310** is axially aligned with lumen **628b**. In this way, optical fibers **301** and **302** may be routed from the fiber optic umbilical **500** and respectively into lumens **628a** and **628b** of the fixture **605c**, and then into the fiber lumens **309** and **310** of the infusion shaft **300c**. Examples of how the fiber **301** and **302** may reside inside the infusion shaft are discussed above.

(256) According to some implementations, a roundabout **630** is located inside the hub tray **601** which is used to route the optical fibers to their destinations in a manner that prevents the fibers from bending beyond their minimum bending radius. For example, fibers **301** and **302** may extend from the umbilical hub connector **502** in a counter-clockwise direction about the roundabout **630** and into their respective lumens **628a** and **628b** of the infusion shaft support fixture **605c**. According to some implementations the radius of curvature of the one or more bends in the roundabout **630** is each greater than the minimum bending radius the optical fibers **301** and **302**. The remainder of the optical fibers may be delivered to infusion shafts **300a** and **300b** in a similar way, but in a clockwise direction about the roundabout **630**.

(257) Upon optical fibers **301** and **302** being placed inside infusion shaft **300c**, mandrels **604a** and **604b** may be introduced respectively into the working lumens **308a** and **308b** through the proximal opening in the proximal connector **304**. The lengths of the mandrels are sufficient for them to pass through the infusion shaft **300c** and through the hub tray **601** until their distal ends reside supported in the mandrel support fixture **612**. When the mandrels **604a** and **604b** are in place, each of their distal ends respectively resides inside the through lumens **614a** and **614b** of the fixture **612** and their proximal ends reside outside the proximal connector **304** of the infusion shaft **300c**. When the optical fibers and mandrels for all of the infusions shafts and main shaft are in place, the assembly is cast as discussed above to form an encapsulated hub that has an appearance similar to the hub **400** shown in FIG. 3.

(258) As discussed above the proximal side **615** of the mandrel support fixture **612** is sloped. The purpose of the slope is to align the proximal side **615** in a plane that is parallel with that of the distal side **624** of the infusion shaft support fixture **605c**. This provides a direct line of sight between the lumens **619a** and **619b** of fixture **605c** with the lumens **614a** and **614b** of fixture **612** to allow the placement of straight rigid mandrels between the respective lumens.

(259) While specific implementations and applications have been illustrated and described, it is to be understood that the invention is not limited to the precise configuration and components disclosed herein. Various modifications, changes, and variations which will be apparent to those skilled in the art may be made in the arrangement, operation, and details of the methods and systems of the present invention disclosed herein without departing from the spirit and scope of the invention.

(260) For example, the disclosure describes in detail various implementations of a CVC and of its individual components. It is appreciated, however, that the disclosed inventive features are applicable to a host of other types of devices inside and outside the medical field. As mentioned above, the apparatus and methods disclosed herein can also be applied to equipment or components of water processing plants, food processing plants, dairies, livestock habitation facilities, etc.

(261) The following clauses disclose in an unlimited way additional implementations, with each clause representing an implementation. Additional implementations are represented by one or more of the implementations of one group or groups of clauses with one or more implementations of another group or groups of clauses. Group A through H clauses are provided.

(262) Group A Clauses:

(263) Clause 1. An assembly comprising: a first optical fiber having a length; a first body having a first receptacle in which at least a portion of the length of the first optical fiber resides.

(264) Clause 2. The assembly according to clause 1, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body.

(265) Clause 3. The assembly according to clause 1, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the first body.

(266) Clause 4. The assembly according to clause 1, wherein the first optical fiber is a radially emitting and end emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body.

(267) Clause 5. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body.

(268) Clause 6. The assembly according to clause 1, further comprising a second optical fiber having a length, the first body having a second receptacle in which at least a portion of the length of the second optical fiber resides.

(269) Clause 7. The assembly according to clause 6, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a first portion of the first body, and the second optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a second portion of the first body.

(270) Clause 8. The assembly according to clauses 1, 3 and 7, wherein at least a portion of the first optical fiber is attached to the first body inside the first receptacle.

(271) Clause 9. The assembly according to clauses 1, 2 and 4, wherein the first optical fiber is not attached to the first body and is slideable inside the first receptacle.

(272) Clause 10. The assembly according to clauses 3, 5 and 7, wherein the first body includes one or more optical surfaces that direct the end emitted bacterial disinfecting light onto the second portion of the first body.

(273) Clause 11. The assembly according to clause 6, wherein the second optical fiber is an imaging fiber.

(274) Clause 12. The assembly according to clauses 2, 3, 6, 9 and 11, wherein the first body is a main shaft of a central venous catheter.

(275) Clause 13. The assembly according to clauses 2-7 and 9-10, wherein the first body is an infusion shaft of a central venous catheter.

(276) Clause 14. The assembly according to clauses 1-13, wherein the first body has a working lumen and an external surface, the external surface comprising a light reflective coating that is configured to reflect at least a portion of the radial emitted disinfecting light into the working lumen.

(277) Clause 15. The assembly according to clauses 1-13, wherein the first body has a working lumen and an external surface, the external surface having disposed thereon a reflective film that is configured to reflect at least a portion of the radial emitted disinfecting light into the working lumen.

(278) Clause 16. The assembly according to clause 15, wherein the film is heat shrunk onto the external surface of the first body.

(279) Clause 17. The assembly according to clauses 1, 2, 4 and 5-7, wherein the first body has a working lumen and a light reflector, the light reflector being embedded in the first body and configured to reflect at least a portion of the radial emitted disinfecting light into the working lumen.

(280) Clause 18. The assembly according to clauses 1, 3 and 5-7, wherein the first body has a working lumen and a light reflector, the light reflector being embedded in the first body and configured to reflect at least a portion of the end emitted disinfecting light into the working lumen.

(281) Clause 19. The assembly according to clause 7, wherein the first body has a working lumen and a light reflector located between the first and second receptacles, the light reflector being embedded in the first body and configured to reflect at least a portion of the radially emitted

disinfecting light into the working lumen.

(282) Clause 20. The assembly according to clauses 1-5, wherein a part of the first body that contains the first and second optical fibers is resistive to being bent beyond a minimum bending radius of the first and second optical fiber.

(283) Clause 21. The assembly according to clauses 3, 5 and 7, wherein the first optical fiber comprises at an end thereof a power density lowering end cap.

(284) Group B Clauses:

(285) Clause 1. An assembly comprising: a first optical fiber having a length; a first body having a receptacle in which at least a first portion of the length of the first optical fiber resides; a second body having formed therein a first channel that houses at least a second portion of the length of the first optical fiber.

(286) Clause 2. The assembly according to clause 1, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body.

(287) Clause 3. The assembly according to clause 1, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the second body.

(288) Clause 4. The assembly according to clause 1, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body and at least a portion of the second body.

(289) Clause 5. The assembly according to clause 1, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the first body.

(290) Clause 6. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body.

(291) Clause 7. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body and at least a portion of the second body.

(292) Clause 8. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body.

(293) Clause 9. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and a portion of the second body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body.

(294) Clause 10. The assembly according to clause 1, further comprising a second optical fiber that has a length, wherein the second body has a second channel that houses at least a portion of the length of the second optical fiber.

(295) Clause 11. The assembly according to clause 10, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body and the second optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the second body.

(296) Clause 12. The assembly according to clause 11, wherein the first optical fiber is configured to radially emit bacterial disinfecting light to disinfect at least a second portion of the second body.

(297) Clause 13. The assembly according to clause 10, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a first portion of the first body, and the second optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a second portion of the first body.

(298) Clause 14. The assembly according to clause 10, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a first portion of the first body, and the second optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a second portion of the first body and a portion of the second body.

(299) Clause 15. The assembly according to clause 10, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a first portion of the first body, and the second optical fiber is also an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the second body.

(300) Clause 16. The assembly according to clause 10, wherein the second optical fiber is configured to end emit bacterial disinfecting light to disinfect at least a portion of the first body.

(301) Clause 17. The assembly according to clause 10, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body, the second optical fiber being a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the second body and to end emit bacterial disinfecting light to disinfect at least a second portion of the second body.

(302) Clause 18. The assembly according to clause 10, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body, the second optical fiber being a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the second body.

(303) Clause 19. The assembly according to clause 10, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body, the second optical fiber being an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the second body.

(304) Clause 20. The assembly according to clause 10, wherein a portion of the second channel overlaps a portion of the first channel.

(305) Clause 21. The assembly according to clause 10, wherein the first body is a flexible structure and the second body is a rigid structure.

(306) Clause 22. The assembly according to clause 1, wherein the first body is a flexible structure and the second body is a rigid structure.

(307) Clause 23. The assembly according to clauses 1-22, wherein at least a portion of the first optical fiber is attached to the first body inside the receptacle.

(308) Clause 24. The assembly according to clauses 1-22, wherein the first optical fiber is not attached to the first body and is slideable inside the receptacle.

(309) Clause 25. The assembly according to clauses 1-24, wherein the at least second portion of the length of the first optical fiber is not held taut along at least a portion of the length of the first channel.

(310) Clause 26. The assembly according to clauses 10-24, wherein the at least second portion of the length of the first optical fiber is not held taut along at least a portion of the length of the first channel and the at least second portion of the length of the second optical fiber is not held taut along at least a portion of the length of the second channel.

(311) Clause 27. The assembly according to clause 10-22, wherein at least a portion of the length of the first channel and at least a portion of the length of the second channel reside at different heights inside the second body.

(312) Clause 28. The assembly according to clause 10-22, wherein the second body comprises a

raised platform that has side walls that at least partially define the first and second channels, the first optical fiber being routed inside the second body in a clockwise direction around the raised platform and the second optical fiber being routed inside the second body in a counter-clockwise direction around the raised platform.

(313) Clause 29. The assembly according to clauses 5 and 13-19, further comprising one or more optical surfaces that direct the end emitted bacterial disinfecting light onto the first body.

(314) Clause 30. The assembly according to clauses 15-17 and 19, further comprising one or more optical surfaces that direct the end emitted bacterial disinfecting light onto the second body.

(315) Clause 31. The assembly according to clauses 1-27, wherein the first body and second body are respectively a main shaft and a hub of a central venous catheter.

(316) Clause 32. The assembly according to clauses 1-27, wherein the first body and second body are respectively an infusion shaft and a hub of a central venous catheter.

(317) Clause 33. The assembly according to clause 31, further comprising an infusion shaft, the infusion shaft and main shaft being fluidly coupled inside the hub by a conduit.

(318) Clause 34. The assembly according to clause 33, wherein at least a portion of one of the first channel and second channel resides above, below or to the side of the conduit.

(319) Clause 35. The assembly according to clauses 3, 5 and 7, wherein the end emitting fiber comprises at an end thereof a power density lowering end cap.

(320) Clause 36. An assembly comprising: an end emitting optical fiber having a length; a body having formed therein a channel that houses at least a portion of the length of the end emitting optical fiber, the end emitting optical fiber configured to end emit bacterial disinfecting light to disinfect at least a portion of the first body.

Group C Clauses:

(321) Clause 1. An assembly comprising: a first optical fiber having a length and a first minimum bending radius; a first body having a receptacle in which at least a first portion of the length of the first optical fiber resides; a second body attached to the first body, the second body having formed therein a first channel that houses at least a second portion of the length of the first optical fiber, the first channel having a length and one or more bends along the length, each of the one or more bends having a radius of curvature that is equal to or greater than the first minimum bending radius of the first optical fiber.

(322) Clause 2. The assembly according to clause 1, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body.

(323) Clause 3. The assembly according to clause 1, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the second body.

(324) Clause 4. The assembly according to clause 1, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body and at least a portion of the second body.

(325) Clause 5. The assembly according to clause 1, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the first body.

(326) Clause 6. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body.

(327) Clause 7. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body and at least a portion of the second body.

(328) Clause 8. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a

first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body.

(329) Clause 9. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and a portion of the second body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body.

(330) Clause 10. The assembly according to clause 1, further comprising a second optical fiber that has a length and a second minimum bending radius, wherein the second body has a second channel that houses at least a portion of the length of the second optical fiber, the second channel having a length and one or more bends along the length, each of the one or more bends of the second channel having a radius of curvature that is equal to or greater than the second minimum bending radius of the second optical fiber.

(331) Clause 11. The assembly according to clause 10, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body and the second optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the second body.

(332) Clause 12. The assembly according to clause 11, wherein the first optical fiber is configured to radially emit bacterial disinfecting light to disinfect at least a second portion of the second body.

(333) Clause 13. The assembly according to clause 10, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a first portion of the first body, and the second optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a second portion of the first body.

(334) Clause 14. The assembly according to clause 10, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a first portion of the first body, and the second optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a second portion of the first body and a portion of the second body.

(335) Clause 15. The assembly according to clause 10, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a first portion of the first body, and the second optical fiber is also an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the second body.

(336) Clause 16. The assembly according to clause 10, wherein the second optical fiber is also configured to end emit bacterial disinfecting light to disinfect at least a portion of the first body.

(337) Clause 17. The assembly according to clause 10, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body, the second optical fiber being a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the second body and to end emit bacterial disinfecting light to disinfect at least a second portion of the second body.

(338) Clause 18. The assembly according to clause 10, wherein the first optical fiber is a radially dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body, the second optical fiber being a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the second body.

(339) Clause 19. The assembly according to clause 10, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body, the second optical fiber being an end emitting fiber that is

configured to end emit bacterial disinfecting light to disinfect at least a portion of the second body.
(340) Clause 20. The assembly according to clause 10, wherein a portion of the second channel overlaps a portion of the first channel.

(341) Clause 21. The assembly according to clause 10, wherein the first body is a flexible structure and the second body is a rigid structure.

(342) Clause 22. The assembly according to clause 1, wherein the first body is a flexible structure and the second body is a rigid structure.

(343) Clause 23. The assembly according to clauses 1-22, wherein at least a portion of the first optical fiber is attached to the first body inside the receptacle.

(344) Clause 24. The assembly according to clauses 1-22, wherein the first optical fiber is not attached to the first body and is slideable inside the receptacle.

(345) Clause 25. The assembly according to clauses 1-24, wherein the at least second portion of the length of the first optical fiber is not held taut along at least a portion of the length of the first channel so that the first optical fiber possesses slack.

(346) Clause 26. The assembly according to clauses 10-24, wherein the at least second portion of the length of the first optical fiber is not held taut along at least a portion of the length of the first channel and the at least second portion of the length of the second optical fiber is not held taut along at least a portion of the length of the second channel.

(347) Clause 27. The assembly according to clause 10-22, wherein at least a portion of the length of the first channel and at least a portion of the length of the second channel reside at different heights inside the second body.

(348) Clause 28. The assembly according to clause 10-22, wherein the second body comprises a raised platform that has side walls that at least partially define the first and second channels, the first optical fiber being routed inside the second body in a clockwise direction around the raised platform and the second optical fiber being routed inside the second body in a counter-clockwise direction around the raised platform.

(349) Clause 29. The assembly according to clauses 5 and 13-19, further comprising one or more optical surfaces through which the end emitted bacterial disinfecting light passes before the bacterial disinfecting light is delivered to the first body.

(350) Clause 30. The assembly according to clauses 15-17 and 19, further comprising one or more optical surfaces through which the end emitted bacterial disinfecting light passes before the bacterial disinfecting light is delivered to the first body.

(351) Clause 31. The assembly according to clauses 1-27, wherein the first body and second body are respectively a main shaft and a hub of a central venous catheter.

(352) Clause 32. The assembly according to clauses 1-27, wherein the first body and second body are respectively an infusion shaft and a hub of a central venous catheter.

(353) Clause 33. The assembly according to clause 32, further comprising an infusion shaft, the infusion shaft and main shaft being fluidly coupled inside the hub by a conduit.

(354) Clause 34. The assembly according to clause 33, wherein at least a portion of one of the first channel or second channel resides above, below or to the side of the tubular conduit.

(355) Clause 35. An assembly comprising: an end emitting optical fiber having a length and a minimum bending radius; a body having formed therein a channel that houses at least a portion of the length of the end emitting optical fiber, the channel having a length and one or more bends along the length, each of the one or more bends having a radius of curvature that is equal to or greater than the minimum bending radius of the end emitting optical fiber, the end emitting optical fiber configured to end emit bacterial disinfecting light to disinfect at least a portion of the first body.

Group D Clauses:

(356) Clause 1. An assembly comprising: a first optical fiber having a length; a first body having a receptacle in which at least a first portion of the length of the first optical fiber resides; a second

body in which a portion of the first body resides, the second body having formed therein a first channel that houses at least a second portion of the length of the first optical fiber; wherein the at least second portion of the length of the first optical fiber is not held taut along at least a portion of the length of the first channel so that the first optical fiber possesses slack.

(357) Clause 2. The assembly according to clause 1, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body.

(358) Clause 3. The assembly according to clause 1, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the second body.

(359) Clause 4. The assembly according to clause 1, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body and at least a portion of the second body.

(360) Clause 5. The assembly according to clause 1, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the first body.

(361) Clause 6. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body.

(362) Clause 7. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body and at least a portion of the second body.

(363) Clause 8. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body.

(364) Clause 9. The assembly according to clause 1, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and a portion of the second body, and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body.

(365) Clause 10. The assembly according to clause 1, further comprising a second optical fiber that has a length and the second body has a second channel that houses at least a portion of the length of the second optical fiber; wherein the at least second portion of the length of the first optical fiber is not held taut along at least a portion of the length of the first channel and the at least second portion of the length of the second optical fiber is not held taut along at least a portion of the length of the second channel.

(366) Clause 11. The assembly according to clause 10, wherein the first optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the first body and the second optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the second body.

(367) Clause 12. The assembly according to clause 11, wherein the first optical fiber is configured to radially emit bacterial disinfecting light to disinfect at least a second portion of the second body.

(368) Clause 13. The assembly according to clause 10, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a first portion of the first body, and the second optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a second portion of the first body.

(369) Clause 14. The assembly according to clause 10, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a first portion of the first body, and the second optical fiber is a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a second portion of the first body and

a portion of the second body.

(370) Clause 15. The assembly according to clause 10, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a first portion of the first body, and the second optical fiber is also an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the second body.

(371) Clause 16. The assembly according to clause 10, wherein the second optical fiber is also configured to end emit bacterial disinfecting light to disinfect at least a portion of the first body.

(372) Clause 17. The assembly according to clause 10, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body, the second optical fiber being a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the second body and to end emit bacterial disinfecting light to disinfect at least a second portion of the second body.

(373) Clause 18. The assembly according to clause 10, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body, the second optical fiber being a radially emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a portion of the second body.

(374) Clause 19. The assembly according to clause 10, wherein the first optical fiber is a dual emitting fiber that is configured to radially emit bacterial disinfecting light to disinfect at least a first portion of the first body and to end emit bacterial disinfecting light to disinfect at least a second portion of the first body, the second optical fiber being an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the second body.

(375) Clause 20. The assembly according to clause 10, wherein a portion of the second channel overlaps a portion of the first channel.

(376) Clause 21. The assembly according to clause 10, wherein the first body is a flexible structure and the second body is a rigid structure.

(377) Clause 22. The assembly according to clause 1, wherein the first body is a flexible structure and the second body is a rigid structure.

(378) Clause 23. The assembly according to clauses 1-22, wherein at least a portion of the first optical fiber is attached to the first body inside the receptacle.

(379) Clause 24. The assembly according to clauses 1-22, wherein the first optical fiber is not attached to the first body and is slideable inside the receptacle.

(380) Clause 25. The assembly according to clause 10-22, wherein at least a portion of the length of the first channel and at least a portion of the length of the second channel reside at different heights inside the second body.

(381) Clause 26. The assembly according to clause 10-22, wherein the second body comprises a raised platform that has side walls that at least partially define the first and second channels, the first optical fiber being routed inside the second body in a clockwise direction around the raised platform and the second optical fiber being routed inside the second body in a counter-clockwise direction around the raised platform.

(382) Clause 27. The assembly according to clauses 5 and 13-19, further comprising one or more optical surfaces through which the end emitted bacterial disinfecting light passes before the bacterial disinfecting light is delivered to the first body.

(383) Clause 28. The assembly according to clauses 15-17 and 19, further comprising one or more optical surfaces through which the end emitted bacterial disinfecting light passes before the bacterial disinfecting light is delivered to the first body.

(384) Clause 29. The assembly according to clauses 1-26, wherein the first body and second body

are respectively a main shaft and a hub of a central venous catheter.

(385) Clause 30. The assembly according to clauses 1-26, wherein the first body and second body are respectively an infusion shaft and a hub of a central venous catheter.

(386) Clause 31. The assembly according to clause 29, further comprising an infusion shaft, the infusion shaft and main shaft being fluidly coupled inside the hub by a conduit.

(387) Clause 32. The assembly according to clause 31, wherein at least a portion of one of the first channel or second channel resides above, below or to the side of the conduit.

(388) Clause 33. An assembly comprising: an end emitting optical fiber having a length; a body having formed therein a channel that houses at least a portion of the length of the end emitting optical fiber, the end emitting optical fiber being configured to end emit bacterial disinfecting light to disinfect at least a portion of the second body; wherein the at least portion of the length of the end emitting optical fiber is not held taut along at least a portion of the length of the channel.

Group E Clauses:

(389) Clause 1. An assembly comprising: an end emitting optical fiber having a length and configured to end emit bacterial disinfecting light; a body having a receptacle in which at least a portion of the length of the end emitting optical fiber resides, the body having a disinfecting target area located therein; one or more surfaces disposed on or in the body onto which the end emitted bacterial disinfecting light is configured to impinge when the end emitting light is activated, the one or more surfaces being configured to alter the trajectory of the disinfecting light so that the disinfecting light is directed toward the disinfecting target area via a disinfecting light pathway.

(390) Clause 2. The assembly according to clause 1, wherein at least one of the one or more surfaces comprises a refractive optical surface.

(391) Clause 3. The assembly according to clause 1, wherein at least one of the one or more surfaces comprises a total internal reflection optical surface.

(392) Clause 4. The assembly according to clause 1, wherein at least one of the one or more surfaces comprises a light reflector.

(393) Clause 5. The assembly according to clause 1, wherein the one or more surfaces comprise a refractive optical surface and a total internal reflection optical surface.

(394) Clause 6. The assembly according to clause 1, wherein the one or more surfaces comprise a refractive optical surface and a light reflector.

(395) Clause 7. The assembly according to clause 1, wherein the one or more surfaces comprise a total internal reflection optical surface and a light reflector.

(396) Clause 8. The assembly according to clause 1, wherein the one or more surfaces comprise a refractive optical surface, a total internal reflection optical surface and a light reflector.

(397) Clause 9. The assembly according to clause 2, wherein the at least one refractive optical surface is configured to collimate the disinfecting light.

(398) Clause 10. The assembly according to clause 2, wherein the least one refractive optical surface is configured to converge the disinfecting light.

(399) Clause 11. The assembly according to clause 2, wherein the least one refractive optical surface is configured to diverge the disinfecting light.

(400) Clause 12. The assembly according to clause 2, wherein the least one refractive optical surface is flat.

(401) Clause 13. The assembly according to clause 2, wherein the least one refractive optical surface has a semi-spherical shape.

(402) Clause 14. The assembly according to clause 2, wherein the least one refractive optical surface has an a-spherical shape.

(403) Clause 15. The assembly according to clause 5, wherein the refractive optical surface precedes the total internal reflection optical surface in the disinfecting light pathway.

(404) Clause 16. The assembly according to clause 6, wherein the refractive optical surface precedes the light reflector in the disinfecting light pathway.

(405) Clause 17. The assembly according to clause 7, wherein the total internal reflection optical surface precedes the light reflector in the disinfecting light pathway.

(406) Clause 18. The assembly according to clause 8, wherein the refractive optical surface precedes the total internal reflection optical surface which precedes the light reflector in the disinfecting light pathway.

(407) Clause 19. The assembly according to clause 2, wherein the end emitting optical fiber comprises a core having an end with there being an index matching material disposed between the end of the core and the refractive optical surface.

(408) Clause 20. The assembly according to clause 19, wherein the index matching material is an adhesive that secures the end emitting optical fiber to the body.

(409) Clause 21. The assembly according to clause 1, wherein the body comprises a part of an infusion shaft of a central venous catheter.

(410) Clause 22. The assembly according to clause 21, wherein the part comprises a proximal connector of the infusion shaft.

(411) Clause 23. The assembly according to clauses 1-22, wherein the end emitting fiber comprises at an end thereof a power density lowering end cap.

(412) Group F Clauses:

(413) Clause 1. An assembly comprising: an elongate shaft having a longitudinal axis and an outer surface; a radially emitting fiber spirally disposed inside the elongate shaft; a plurality of through holes located between the radially emitting fiber and the outer surface, the through holes being longitudinally spaced-apart from one another; a laser optically coupled with the radially emitting fiber; a controller operatively coupled to the laser and configured to cause the laser to deliver pulsed light with a pulsed width into the radially emitting fiber, a longitudinal distance between adjacent through holes being such that the pulse width is shorter than a time for the light to travel between the adjacent through holes.

(414) Clause 2. The assembly according to clause 1, further comprising a back reflectance detector that is optically coupled to the radially emitting fiber and configured to receive back reflected light pulses from the radially emitting fiber.

(415) Clause 3. The assembly according to clause 2, wherein the light source is a laser that is optically coupled to the radially emitting fiber and configured to deliver the pulsed light to the radially emitting fiber via a mirror.

(416) Clause 4. The assembly according to clause 3, wherein the mirror is also configured to direct the back reflected light pulses to the back reflectance detector.

(417) Clause 5. The assembly according to clause 2, wherein the back reflectance detector is configured to generate data based upon the back reflected light pulses received in the back reflectance detector.

(418) Clause 6. The assembly according to clause 5, wherein the back reflectance detector comprises means for comparing the data with baseline data to determine if an unwanted substance exists on the outer surface of the elongate shaft.

(419) Clause 7. The assembly according to clause 5, wherein the back reflectance detector comprises means for processing the data to form an image that can be used to determine if an unwanted substance exists on the outer surface of the elongate shaft.

(420) Clause 8. The assembly according to clause 6, wherein the unwanted substance is a biofilm or a clot.

(421) Clause 9. The assembly according to clause 1, wherein the elongate shaft comprises a wall that extends between an inner surface and the outer surface, the inner surface defining a lumen that runs along a length of the elongate shaft.

(422) Clause 10. The assembly according to clause 1, wherein the radially emitting fiber is fixed inside the elongate shaft.

(423) Clause 11. The assembly according to clause 1, wherein the radially emitting fiber is

disposed inside a spiral lumen inside a length of the elongate shaft, the radially emitting fiber being slideable inside the lumen.

(424) Clause 12. The assembly according to clause 1 comprising a single radially emitting fiber.

(425) Clause 13. The assembly according to clause 1, wherein the radially emitting fiber is located nearer the outer surface of the elongate shaft than to the longitudinal axis of the elongate shaft.

(426) Clause 14. The assembly according to clause 1, wherein the laser is configured to emit red light.

(427) Clause 15. A method of determining the presence of an unwanted substance on the outer surface of an elongate shaft, the elongate shaft including a longitudinal axis, an outer surface, a radially emitting fiber spirally disposed inside the elongate shaft, and a plurality of through holes located between the radially emitting fiber and the outer surface, the through holes being longitudinally spaced-apart from one another, the method comprising: emitting pulsed light having a pulsed width from the radially emitting fiber so that at least a portion of the pulsed light is delivered through the plurality of through holes; sequentially receiving in the radially emitting fiber through the plurality of through holes a plurality of back reflected light pulses; delivering the plurality of back reflected light pulses to a back reflective detector; generating data that correlates with the plurality of back reflected light pulses.

(428) Clause 16. The method according to clause 15, further comprising comparing the data with baseline data to determine if an unwanted substance exists on the outer surface of the elongate shaft.

(429) Clause 17. The method according to clause 15, further comprising means for processing the data to form an image that can be used to determine if an unwanted substance exists on the outer surface of the elongate shaft.

(430) Clause 18. The method according to clause 15, wherein the elongate shaft has a proximal end and a distal end, and the plurality of through holes includes a proximal-most through hole, a distal-most through hole and one or more through holes dispersed there between, the plurality of back reflected pulses being delivered to the back reflective detector sequentially beginning with the back reflected pulse received in the radially emitting fiber through the proximal-most through hole and ending with the back reflected pulse received in the radially emitting fiber through the distal-most through hole.

(431) Clause 19. The method according to clause 15, further comprising delivery light from a laser to the radially emitting fiber via a mirror and delivering the plurality of back reflected pulses to the back reflectance detector via the mirror.

(432) Clause 20. The method according to clause 19, wherein the light is red light.

(433) Group G Clauses:

(434) Clause 1. A hub comprising: a first optical fiber, the first optical fiber having a length and a first minimum bending radius; a first channel that houses at least a portion of the length of the first optical fiber, the first channel having a length and one or more bends along the length, each of the one or more bends having a radius of curvature that is equal to or greater than the first minimum bending radius of the first optical fiber.

(435) Clause 2. The hub according to clause 1, wherein the first optical fiber is a radially emitting fiber that is configured to emit bacterial disinfecting light to disinfect at least a portion of the hub.

(436) Clause 3. The hub according to clause 1, wherein the first optical fiber is an end emitting fiber that is configured to emit bacterial disinfecting light to disinfect at least a portion of the hub.

(437) Clause 4. The hub according to clause 3, further comprising a proximal end portion of a main shaft, the main shaft including a working lumen that has a proximal opening located inside the hub, the end emitting fiber being configured to emit bacterial disinfecting light to disinfect at least a portion of the proximal end portion of the main shaft.

(438) Clause 5. The hub according to clause 1, further comprising: a distal end portion of an infusion shaft, the infusion shaft including a working lumen that has a distal opening located inside

the hub; a proximal end portion of a main shaft, the main shaft including a working lumen that has a proximal opening located inside the hub; a conduit that fluidly couples the distal opening of the working lumen of the infusion shaft to the proximal opening of the working lumen of the main shaft, the conduit having a length; at least a portion of the length of the first channel that houses the first optical fiber being located above, below or to the side of at least a portion of the length of the conduit.

(439) Clause 6. The hub according to clause 5, wherein the first optical fiber is a radially emitting fiber that is configured to emit a bacterial disinfecting light and to expose the conduit to the bacterial disinfecting light.

(440) Clause 7. The hub according to clause 1, further comprising: a distal end portion of an infusion shaft, the infusion shaft including a working lumen that has a distal opening located inside the hub; a proximal end portion a main shaft, the main shaft including a working lumen that has a proximal opening located inside the hub; a conduit that fluidly couples the distal opening of the working lumen of the infusion shaft to the proximal opening of the working lumen of the main shaft, the conduit having a length; wherein the first optical fiber is an end emitting fiber that is configured to emit bacterial disinfecting light to disinfect at least a portion of the hub.

(441) Clause 8. The hub according to clause 6, wherein the conduit comprises a tubular body that is transparent or translucent to the bacterial disinfecting light.

(442) Clause 9. The hub according to clause 1, further comprising: a distal end portion of an infusion shaft, the infusion shaft including a working lumen that has a distal opening located inside the hub; a proximal end portion a main shaft, the main shaft including a working lumen that has a proximal opening located inside the hub; a conduit that fluidly couples the distal opening of the working lumen of the infusion shaft to the proximal opening of the working lumen of the main shaft, the conduit having a length; at least a portion of the length of the first channel that houses the first optical fiber being located above or below at least a portion of a length of the working lumen located in the distal portion of the infusion shaft.

(443) Clause 10. The hub according to clause 9, wherein the first optical fiber is a radially emitting fiber that is configured to emit a bacterial disinfecting light and to expose the at least portion of the length of the working lumen of the infusion shaft to the bacterial disinfecting light.

(444) Clause 11. The hub according to clause 9, wherein the first optical fiber is an end emitting fiber that is configured to emit a bacterial disinfecting light and to expose the at least portion of the length of the working lumen of the infusion shaft to the bacterial disinfecting light.

(445) Clause 12. The hub according to clause 10, wherein the at least portion of the length of the working lumen of the infusion shaft resides inside a tubular body that is transparent or translucent to the bacterial disinfecting light.

(446) Clause 13. The hub according to clause 11, wherein the at least portion of the length of the working lumen of the infusion shaft resides inside a tubular body, the tubular body having an opening through which the bacterial disinfecting light is configured to propagate into the working lumen.

(447) Clause 14. The hub according to clause 11, further comprising one or more optical surfaces through which the bacterial disinfecting light emitted by the end emitting fiber passes before the bacterial disinfecting light is delivered to the at least portion of the length of the working lumen of the infusion shaft.

(448) Clause 15. The hub according to clause 5, wherein the distal end portion of the infusion shaft includes a fiber lumen that contains at least a portion of the first optical fiber.

(449) Clause 16. The hub according to clause 15, wherein the first optical fiber is a radially emitting fiber that is configured to emit bacterial disinfecting light.

(450) Clause 17. The hub according to clause 15, wherein the first optical fiber is an end emitting fiber that is configured to emit bacterial disinfecting light.

(451) Clause 18. The hub according to clause 9, wherein the proximal end portion of the main shaft

includes a fiber lumen that contains at least a portion of the first optical fiber.

(452) Clause 19. The hub according to clause 18, wherein the first optical fiber is a radially emitting fiber that is configured to emit bacterial disinfecting light.

(453) Clause 20. The hub according to clause 18, wherein the first optical fiber is an end emitting fiber that is configured to emit bacterial disinfecting light.

(454) Clause 21. The hub according to clause 5, further comprising: a second optical fiber, the second optical fiber having a length and a second minimum bending radius; a second channel that houses at least a portion of the length of the second optical fiber, the second channel having a length and one or more bends along the length, each of the one or more bends having a radius of curvature that is equal to or greater than the second minimum bending radius of the second optical fiber; at least a portion of the length of the second channel that houses the second optical fiber being located above or below at least a portion of a length of the working lumen located in the distal portion of the infusion shaft.

(455) Clause 22. The hub according to clause 21, wherein at least a portion of the length of the first channel and at least a portion of the length of the second channel overlap.

(456) Clause 23. The hub according to clause 21, wherein at least a portion of the length of the second channel resides vertically above at least a portion of the length of the first channel.

(457) Clause 24. The hub according to clause 21, wherein each of the first and second optical fibers is a radially emitting fiber that is configured to emit bacterial disinfecting light.

(458) Clause 25. The hub according to clause 21, wherein each of the first and second optical fibers is an end emitting fiber that is configured to emit bacterial disinfecting light.

(459) Clause 26. The hub according to clause 21, wherein the first optical fiber is a radially emitting fiber that is configured to emit bacterial disinfecting light and the second optical fiber is an end-emitting fiber that is configured to emit bacterial emitting light.

(460) Clause 27. The hub according to clause 24, wherein the proximal end portion of the main shaft includes a fiber lumen that contains at least a portion of the second optical fiber.

(461) Clause 28. The hub according to clause 24, wherein the distal end portion of the infusion shaft includes a fiber lumen that contains at least a portion of the first optical fiber.

(462) Clause 29. The hub according to clause 27, wherein the distal end portion of the infusion shaft includes a fiber lumen that contains at least a portion of the first optical fiber.

(463) Clause 30. The hub according to clause 1, further comprising: a distal end portion of an infusion shaft, the infusion shaft including a working lumen that has a distal opening located inside the hub; a proximal end portion a main shaft, the main shaft including a working lumen that has a proximal opening located inside the hub, the proximal end portion of the main shaft having a fiber lumen; a conduit that fluidly couples the distal opening of the working lumen of the infusion shaft to the proximal opening of the working lumen of the main shaft, the conduit having a length; wherein the first optical fiber is an imaging fiber, at least a portion of a length of the imaging fiber residing inside the fiber lumen.

(464) Clause 31. The hub according to clause 1, wherein the first optical fiber is not held taut along at least a portion of the length of the first channel.

(465) Clause 32. The hub according to clause 5, wherein the first optical fiber is not held taut along at least a portion of the length of the first channel.

(466) Clause 33. The hub according to clause 21, wherein the first optical fiber is not held taut along at least a portion of the length of the first channel and the second optical fiber is not held taut along at least a portion of the length of the second channel.

(467) Group H Clauses:

(468) Clause 1. An assembly comprising: a polymeric body having walls that define a working lumen and a fiber lumen, the polymeric body having a first part in which the working lumen resides and a second part in which the fiber lumen resides, the first part having a first external shape and the second part having a second external shape that is different from the first shape, the polymeric

body having an external profile defined by a perimeter of the first and second parts; and a clamp transitional between an open position and a closed position, in the closed position the clamp applies a force on the second part of the polymeric body to cause a deformation of the first part to cause a full or partial closure of the working lumen, the clamp having a proximal wall and a distal wall, at least one of the proximal and distal walls having a through opening that has an internal profile that is substantially similar to the external profile of the polymeric body, at least a portion of the polymeric body residing in the through opening so that the polymeric body is prevented from rotating in the clamp.

(469) Clause 2. The assembly according to clause 1, further comprising an optical fiber residing in the fiber lumen.

(470) Clause 3. The assembly according to clause 1, wherein the working lumen is defined by a top wall having a first thickness, a bottom wall having a second thickness and first and second side walls that each have a third thickness, the first and second side walls being located between the top and bottom walls, the third thickness being less than both the first and second thicknesses, the dimensional characteristics of the top wall, bottom wall and first and second side walls result in a bending of the first and second side walls when a downward force is applied to the top wall by the clamp, the bending allowing an inner surface of the top wall to contact an inner surface of the bottom wall to effectuate the full or partial closure of the working lumen.

(471) Clause 4. The assembly according to clause 1, wherein a wall segment that defines at least in part the fiber lumen protrudes into the working lumen.

(472) Clause 5. The assembly according to clause 3, wherein a wall segment that defines at least in part the fiber lumen protrudes into the working lumen.

(473) Clause 6. The assembly according to clause 5, wherein the inner surface of the top wall includes a recess that is configured to mate with the protruding wall segment.

(474) Clause 7. The assembly according to clause 2, wherein the first optical fiber is a radially emitting fiber.

(475) Clause 8. The assembly according to clause 2, wherein the first optical fiber is an end emitting fiber.

(476) Clause 9. The assembly according to clause 1, wherein the clamp includes an upper pad and a lower pad, the lower pad including a groove in which at least a portion of the first part of the polymeric body resides.

(477) Clause 10. The assembly according to clause 9, wherein the first part and the groove are configured to inhibit a rotating of the polymeric body in the clamp.

(478) Clause 11. The assembly according to clause 9, wherein when clamp is in the open position the upper pad does not press against the polymeric body and when the clamp is in the closed position the upper pad presses downward on the polymeric body, the clamp being continuously urged in the open direction.

(479) Clause 12. The assembly according to clause 11, wherein the clamp includes a latch assembly to hold the clamp in the closed position.

(480) Clause 13. The assembly according to clause 9, wherein the clamp includes a base onto which the lower pad is attached and an arm onto which the upper pad is attached, the arm being joined to the base by the proximal wall.

(481) Clause 14. The assembly according to clause 13, wherein the proximal wall includes a hinge.

(482) Clause 15. The assembly according to clause 1, wherein the polymeric body is an infusion shaft of a central venous catheter.

(483) Clause 16. The assembly according to clause 1, wherein each of the proximal and distal walls have a through opening that has an internal profile that is substantially similar to the external profile of the polymeric body, at least a portion of the polymeric body residing in each of the through openings so that the polymeric body is prevented from rotating in the clamp.

(484) Clause 17. The assembly according to clause 11, wherein the clamp further comprises a stop

to limit the amount by which the upper pad can be pressed against the polymeric body.

(485) Clause 18. The assembly according to clause 13, wherein a bottom of the base includes one or more slip resistant grips.

(486) Clause 19. The assembly according to clause 13, wherein a top of the arm includes one or more slip resistant grips.

(487) Clause 20. The assembly according to clause 15 wherein the infusion shaft includes a proximal connector, the clamp being directly attached to the proximal connector.

(488) Clause 21. An assembly comprising: a polymeric body having walls that define a first working lumen, a second working lumen and a fiber lumen, the fiber lumen being located inside a septum that separates the first and second working lumens, the polymeric body having a first part in which the first working lumen, second working lumen and fiber lumen reside, the polymeric body having a second part that protrudes radially from the first part, the first part having a first external shape and the second part having a second external shape that is different from the first shape, the polymeric body having an external profile defined by a perimeter of the first and second parts; and a clamp that is transitional between an open position and a closed position, in the closed position the clamp applies a force on the second part of the polymeric body to cause a deformation of the first part to cause a full or partial closure of at least one of the first and second working lumens, the clamp having a proximal wall and a distal wall, at least one of the proximal and distal walls having a through opening that has an internal profile that is substantially similar to the external profile of the polymeric body, at least a portion of the polymeric body residing in the through opening so that the polymeric body is prevented from rotating in the clamp.

(489) Clause 22. The assembly according to clause 21, further comprising an optical fiber residing in the fiber lumen.

(490) Clause 23. The assembly according to clause 21, wherein a part of the septum that defines the fiber lumen protrudes into one or both of the first and second working lumens.

(491) Clause 24. The assembly according to clause 22, wherein the first optical fiber is a radially emitting fiber.

(492) Clause 25. The assembly according to clause 22, wherein the first optical fiber is an end emitting fiber.

(493) Clause 26. The assembly according to clause 21, wherein the clamp is continuously urged in the open direction and includes a latch assembly to hold the clamp in the closed position.

(494) Clause 27. The assembly according to clause 21, wherein the clamp includes a base and an arm that is joined to the base by the proximal wall the through opening of the proximal wall having an internal profile that is substantially similar to the external profile of the polymeric body.

(495) Clause 28. The assembly according to clause 27, wherein the proximal wall includes a hinge.

(496) Clause 29. The assembly according to clause 21, wherein the polymeric body is an infusion shaft of a central venous catheter.

(497) Clause 30. The assembly according to clause 21, wherein each of the proximal and distal walls have a through opening that has an internal profile that is substantially similar to the external profile of the polymeric body, at least a portion of the polymeric body residing in each of the through openings so that the polymeric body is prevented from rotating in the clamp.

Claims

1. A central venous catheter comprising: a main shaft having a proximal end portion and a distal end portion, the distal end portion configured for placement inside a patient, the main shaft having a working lumen; a first infusion shaft having a proximal end portion and a distal end portion, the first infusion shaft having a working lumen; a first optical fiber having a length and being configured to emit bacterial disinfecting light; and a hub having a proximal end and a distal end, the distal end portion of the first infusion shaft being coupled to the main shaft by a first conduit

located inside the hub, the first conduit fluidly connecting the working lumen of the first infusion shaft with the working lumen of the main shaft, the hub having a first channel that houses at least a portion of the length of the first optical fiber, the first channel being located outside the conduit and not forming a part of the main shaft, nor a part of the infusion shaft.

2. The central venous catheter according to claim 1, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the hub.

3. The central venous catheter according to claim 2, wherein the first optical fiber is configured to end emit bacterial disinfecting light to disinfect at least a portion of the first conduit.

4. The central venous catheter according to claim 3, wherein the first conduit comprises a tubular body that is transparent or translucent to the bacterial disinfecting light.

5. The central venous catheter according to claim 1, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the first conduit.

6. The central venous catheter according to claim 5, wherein the first conduit comprises a tubular body that is transparent or translucent to the bacterial disinfecting light.

7. The central venous catheter according to claim 1, wherein the first conduit comprises a tubular body that is transparent or translucent to the bacterial disinfecting light.

8. The central venous catheter according to claim 1, wherein the first channel includes a first segment at a first elevation and a second segment at a second elevation different than the first elevation.

9. The central venous catheter according to claim 1, further comprising a second optical fiber having a length and being configured to emit bacterial disinfecting light; and a second channel located inside the hub that houses at least a portion of the length of the second optical fiber.

10. The central venous catheter according to claim 9, wherein each of the first optical fiber and the second optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light.

11. The central venous catheter according to claim 9, wherein each of the first optical fiber and the second optical fiber is an end emitting fiber that is each configured to end emit bacterial disinfecting light to disinfect at least a portion of the hub.

12. The central venous catheter according to claim 9, wherein the hub comprises a raised platform that has a side wall, at least a portion of the length of the second optical fiber being routed inside the hub at least partially around the raised platform.

13. The central venous catheter according to claim 12, wherein at least a portion of the length of each of the first optical fiber and the second optical fiber is routed inside the hub at least partially around the raised platform.

14. The central venous catheter according to claim 9 further comprising: a second infusion shaft having a proximal end portion and a distal end portion, the second infusion shaft having a working lumen; the distal end portion of the second infusion shaft being coupled to the main shaft by a second conduit located inside the hub, the second conduit fluidly connecting the working lumen of the second infusion shaft with the working lumen of the main shaft, the second channel being located outside the second conduit and not forming a part of the main shaft, nor a part of the infusion shaft.

15. The central venous catheter according to claim 14, wherein each of the first optical fiber and the second optical fiber is an end emitting fiber that is each configured to end emit bacterial disinfecting light to disinfect at least a portion of the hub.

16. The central venous catheter according to claim 14, wherein the first optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the first conduit, and the second optical fiber is an end emitting fiber that is configured to end emit bacterial disinfecting light to disinfect at least a portion of the second conduit.

17. The central venous catheter according to claim 16, wherein each of the first conduit and the second conduit comprises a tubular body that is transparent or translucent to the bacterial disinfecting light.

18. The central venous catheter according to claim 9, wherein each of the first optical fiber and the second optical fiber is an end emitting fiber that is configured to end emit bacterial emitting light.

19. The central venous catheter according to claim 1, wherein the hub comprises a raised platform that has a side wall that at least partially defines the first channel.

20. The central venous catheter according to claim 19, wherein at least a portion of the length of the first optical fiber is routed inside the hub at least partially around the raised platform.
